

Nuclear Magnetic Resonance

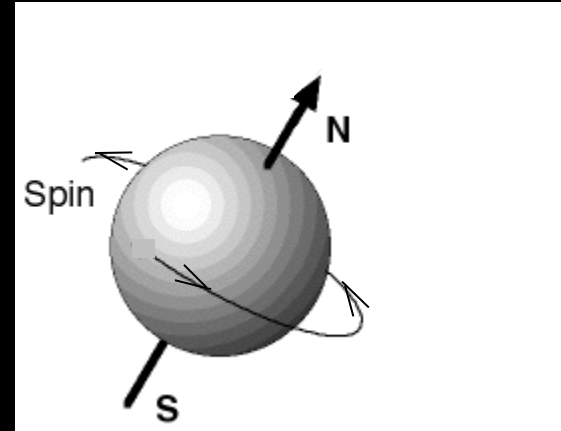
By 1946 Bloch and Purcell captured the essence of the behavior of a nucleus with a non-zero magnetic moment situated in a magnetic field.

Although quantum physics was needed to formulate the experimental and theoretical results, the final impact rests on the very simple relationship given by the Larmor equation relating the frequency of rotation or precession frequency (ω) and the local magnetic field (B).

Proton (^1H)

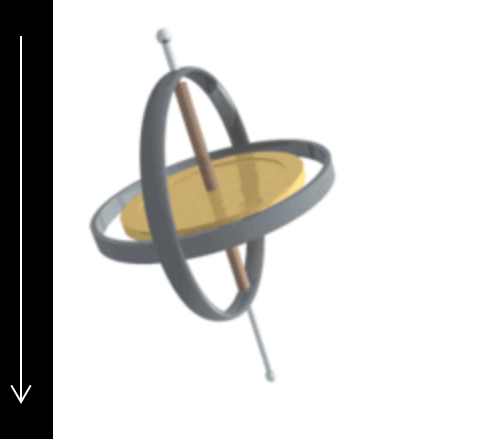
Protons have

- Mass
- Charge
- Spin $\frac{1}{2}$ (angular momentum)
- Magnetic moment



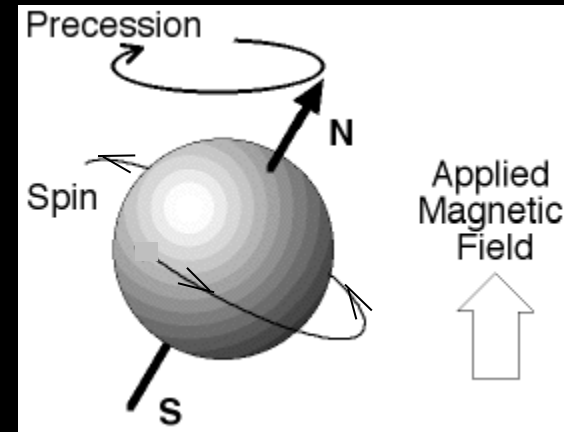
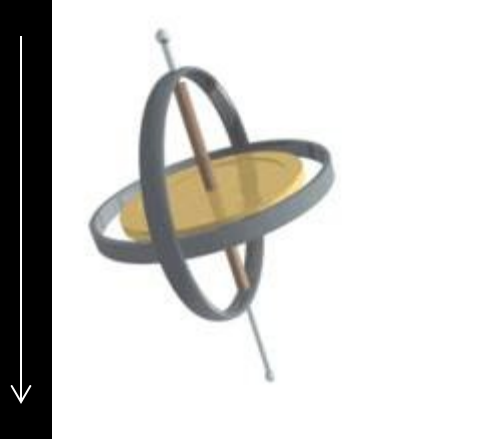
Proton (^1H) - Precession

Earth's
gravitational
field



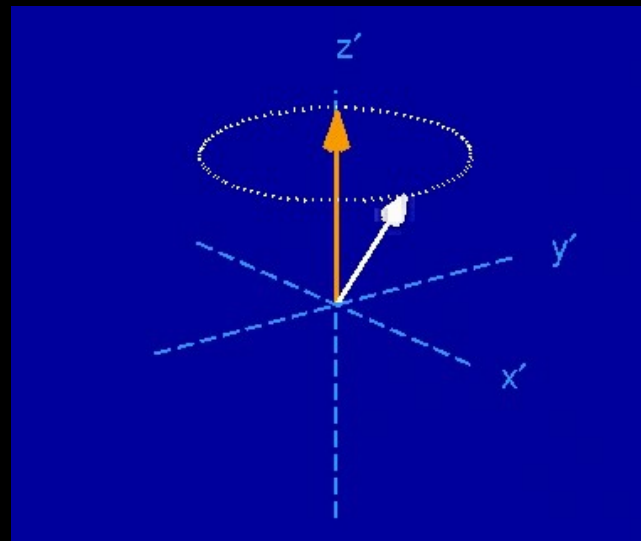
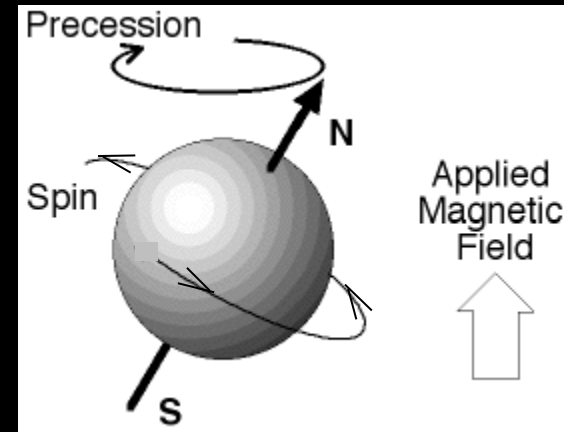
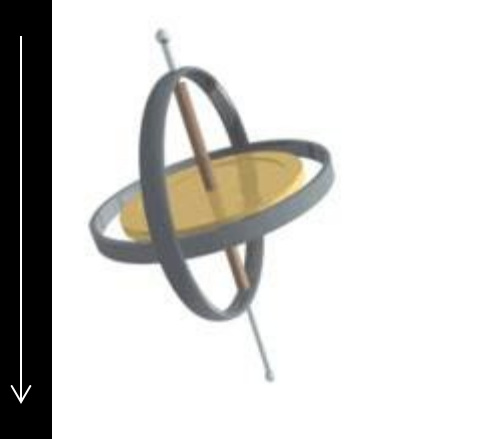
Proton (^1H) - Precession

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Proton (^1H) - Precession

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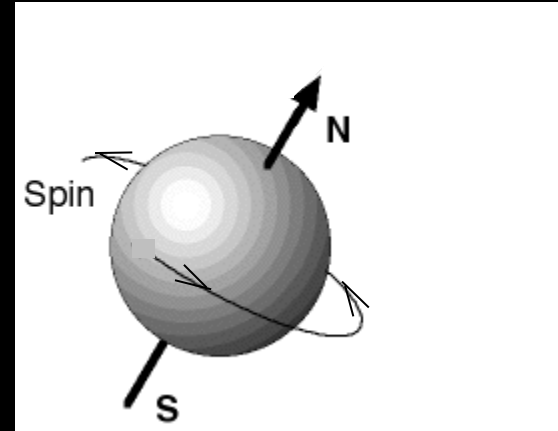


Nuclear
Precession

Proton (^1H) - Precession

Protons have

- Mass
- Charge
- Spin $\frac{1}{2}$ (angular moment)
- Magnetic moment



The spin angular moment, J , and the Magnetic moment, μ are vectorially aligned (parallel or anti-parallel) and hence are related by a proportionality scalar value: γ

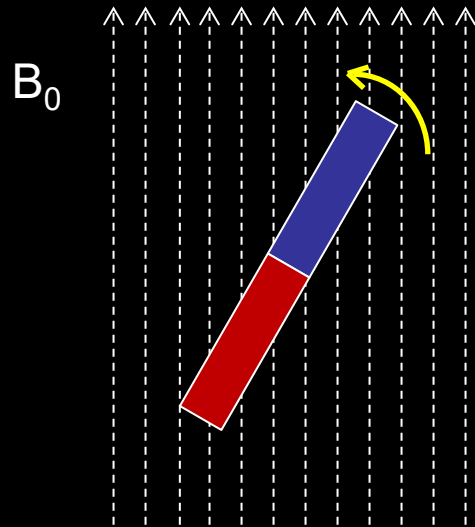
$$\mu = \gamma J$$

$$\gamma \sim \frac{q}{2m}$$

γ - Gyromagnetic ratio

The Larmor Equation

Equation of motion for a single proton



Torque

$$N = \mu \times B$$

$$N = \frac{dJ}{dt}$$

Rate of change of angular momentum

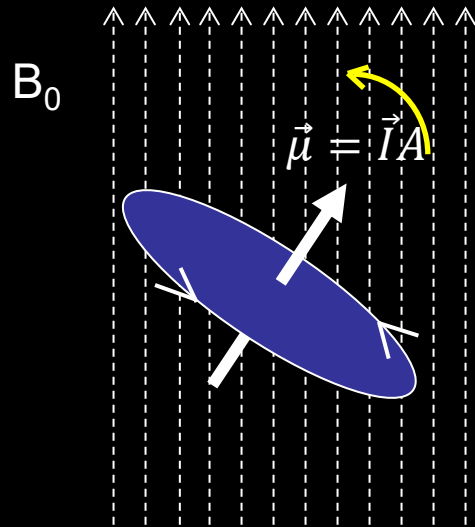
$$\frac{dJ}{dt} = \mu \times B$$

$$\mu = \gamma \cdot J$$

$$\frac{d\mu}{dt} = \gamma \cdot \mu \times B$$

The Larmor Equation

Equation of motion for a single proton



Torque

$$N = \mu \times B$$

$$N = \frac{dJ}{dt}$$

Rate of change of angular momentum

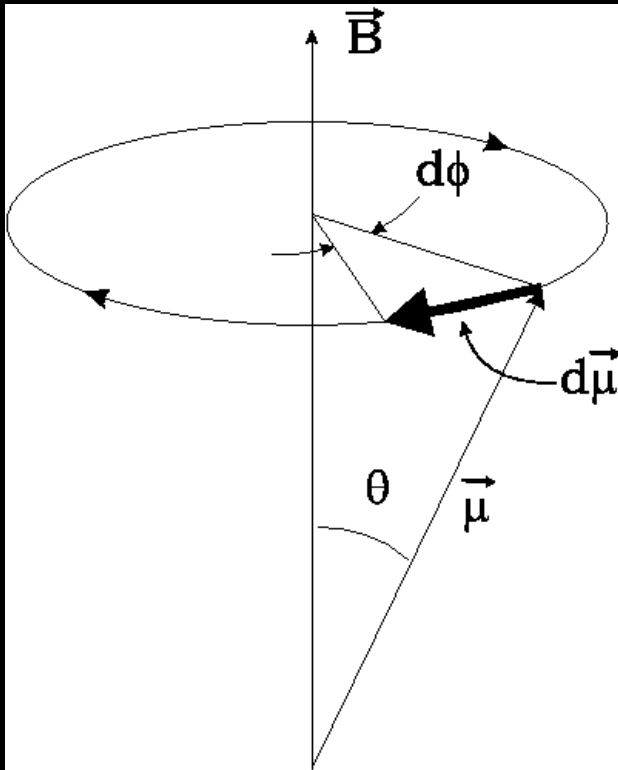
$$\frac{dJ}{dt} = \mu \times B$$

$$\mu = \gamma \cdot J$$

$$\frac{d\mu}{dt} = \gamma \cdot \mu \times B$$

The Larmor Equation

Equation of motion for a single proton



$$\frac{d\vec{\mu}}{dt} = \gamma \cdot \vec{\mu} \times \vec{B}$$

$$|d\vec{\mu}| = \gamma \cdot |\vec{\mu}| \cdot |\vec{B}| \cdot \sin\theta \cdot dt$$

$$|d\vec{\mu}| = |\vec{\mu}| \cdot \sin\theta \cdot |d\phi|$$

$$|\vec{\mu}| \cdot \sin\theta \cdot \frac{d\phi}{dt} = \gamma \cdot |\vec{\mu}| \cdot |\vec{B}| \cdot \sin\theta$$

$$\left| \frac{d\phi}{dt} \right| = \gamma \cdot |\vec{B}|$$

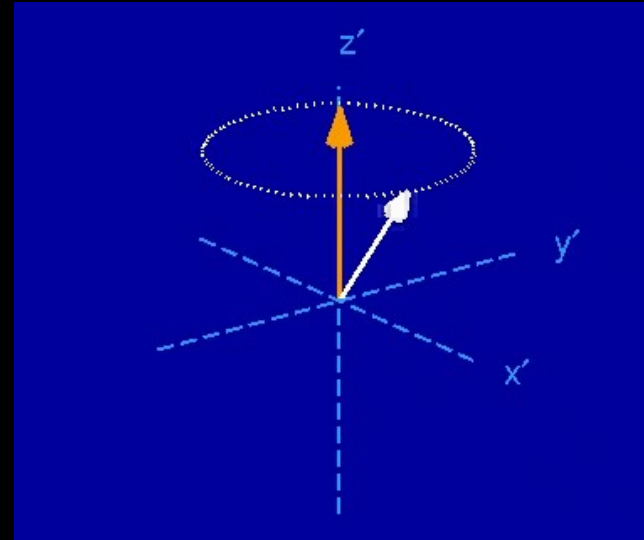
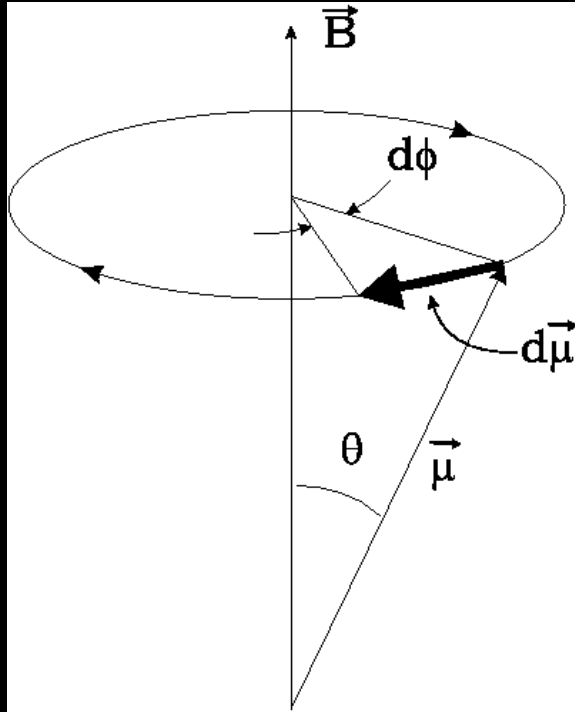
$$\frac{d\phi}{dt} = -\omega$$

Clockwise rotation

$$\vec{\omega}_0 = -\gamma \cdot \vec{B}_0$$

The Larmor Equation

$$\omega_0 = -\gamma \cdot B_0$$



Precession frequency is given by the Larmor equation

The precession frequency is linearly proportional to the applied magnetic field B_0

The Larmor Equation

$$\omega_0 = -\gamma \cdot B_0$$

Gyromagnetic ratio, γ , determines the precession frequency of a nucleus in an external magnetic field

$$\gamma \longrightarrow \frac{\text{radians}}{\text{sec} \cdot \text{Tesla}} \longrightarrow \frac{\text{Hz}}{\text{Tesla}}$$

γ , is a nuclear property and has unique values for different nuclei

The Larmor Equation

$$\omega_0 = -\gamma \cdot B_0$$

For ^1H ,

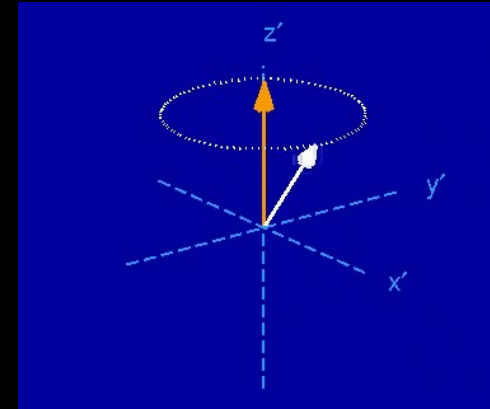
$$\gamma \longrightarrow 2.675 \times 10^8 \frac{\text{radians}}{\text{sec} \cdot \text{Tesla}} \longrightarrow 42.6 \times 10^6 \frac{\text{Hz}}{\text{Tesla}}$$

Earth's magnetic field is 50 microTesla

At what frequency are the Hydrogen protons in our body precessing at this very moment?

The rotating reference frame

$$\frac{d\mu}{dt} = \gamma \cdot \mu \times B$$

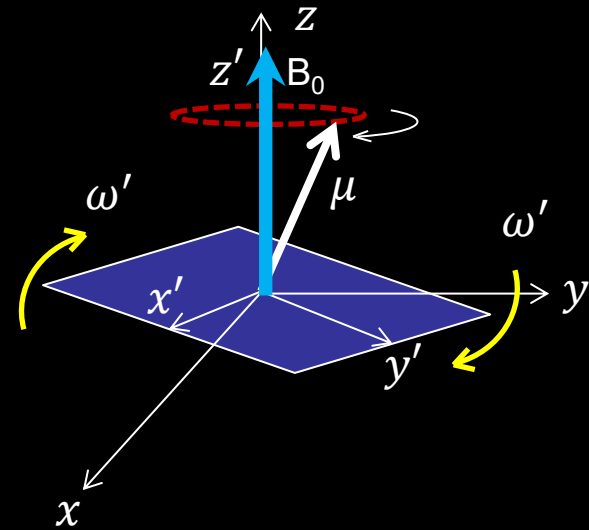
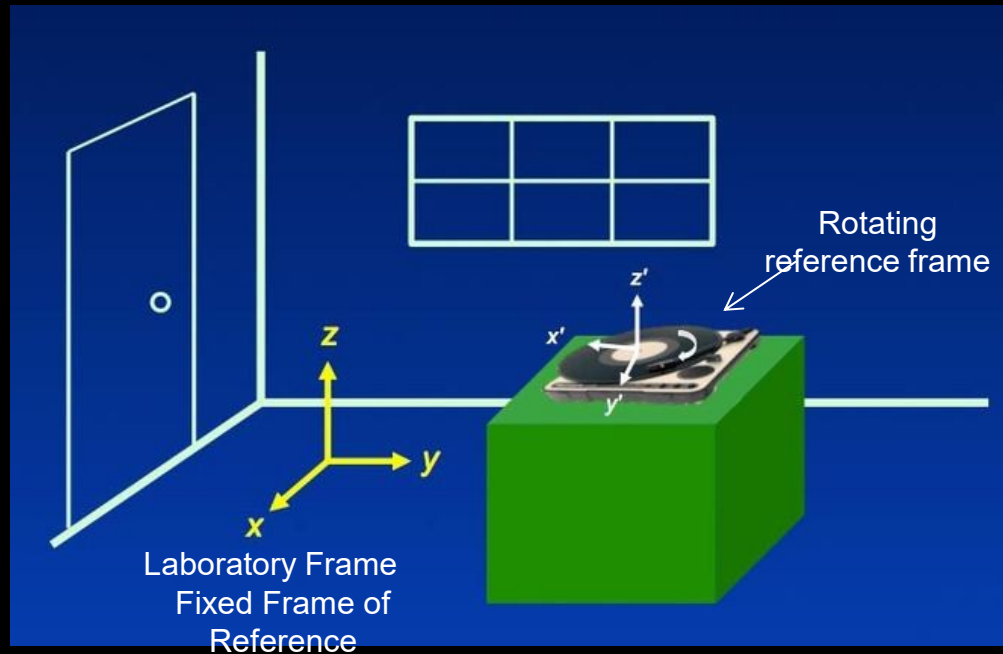


Laboratory frame of reference

All protons, precessing at Larmor Frequency

Since our goal in MR imaging experiment is to manipulate the spin magnetization for our imaging purpose, it is useful to look at the spin motion in a simpler reference frame...

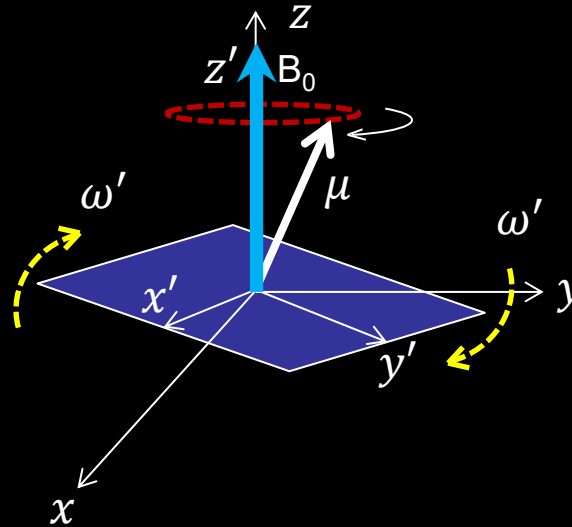
The rotating reference frame



The rotating reference frame

$$\frac{d\mu}{dt} = \gamma \cdot \mu \times B_z$$

$$z \equiv z'$$



$$\left(\frac{d\mu}{dt}\right)' = \gamma \cdot \mu \times \left(B_z + \frac{\omega'}{\gamma}\right)$$

$$\omega' = \omega_0 = -\gamma \cdot B$$

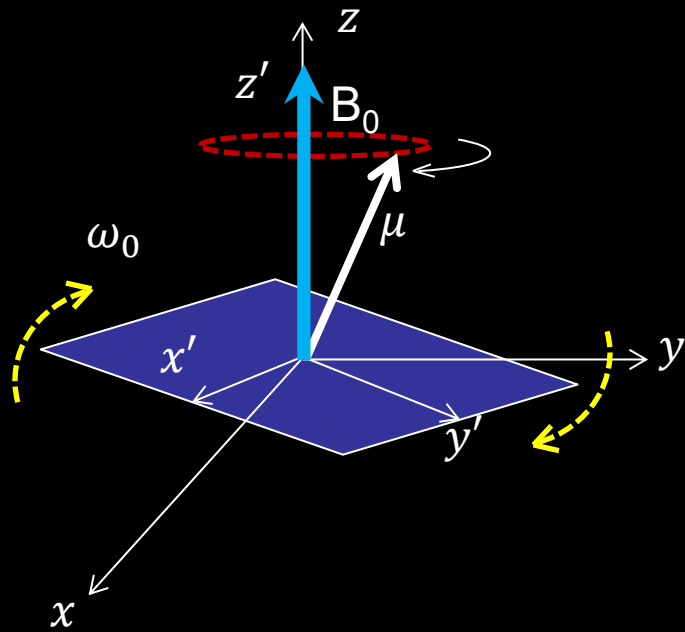
condition of Resonance

$$\left(\frac{d\mu}{dt}\right)' = \gamma \cdot \mu \times \left(B_z + \frac{-\gamma \cdot B_z}{\gamma}\right)$$

$$\left(\frac{d\mu}{dt}\right)' = \gamma \cdot \mu \times (0) = 0$$

No effect of B_0 in an
on- resonance
rotating reference
frame

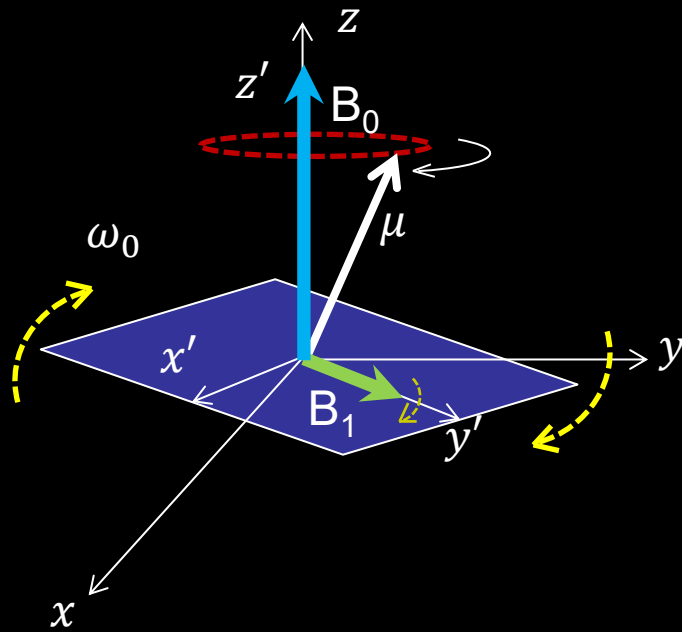
Tipping the Magnetization



Tipping the Magnetization

A rotating B_1 field - ON

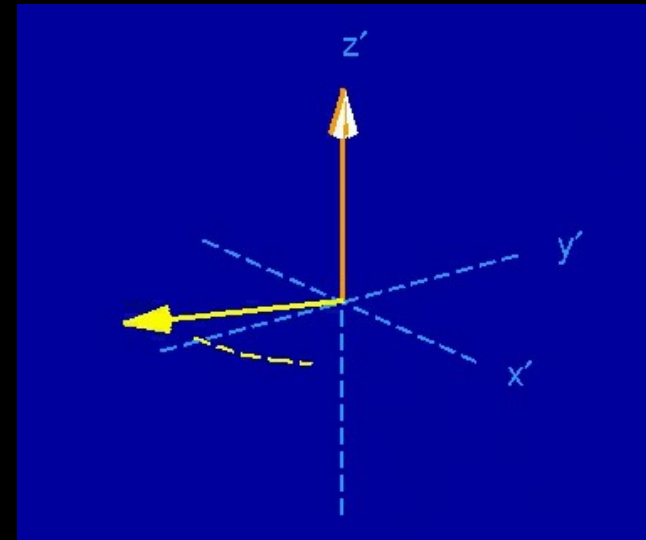
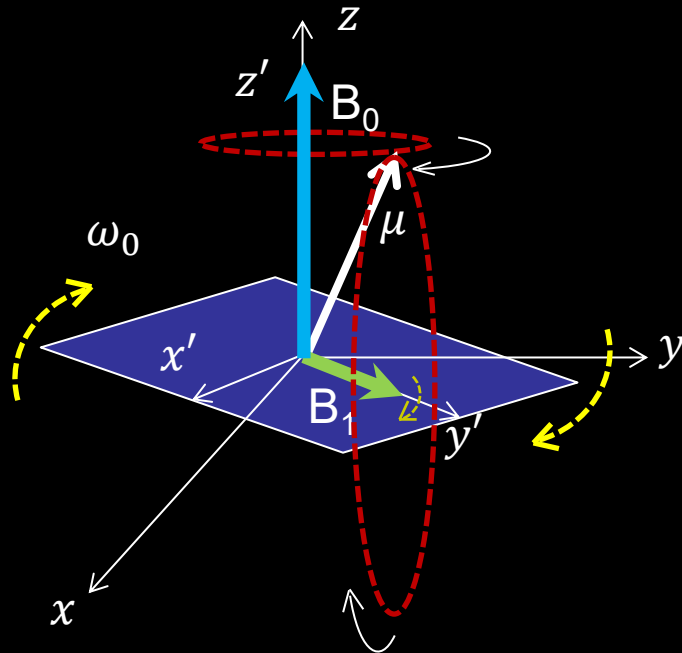
$$\left(\frac{d\mu}{dt}\right)' = \gamma \cdot \mu \times (B_{1x'})$$



Tipping the Magnetization

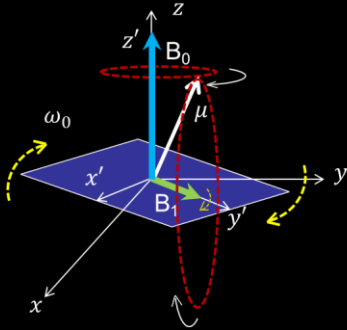
A rotating B_1 field - ON

$$\left(\frac{d\mu}{dt}\right)' = \gamma \cdot \mu \times (B_{1y'})$$



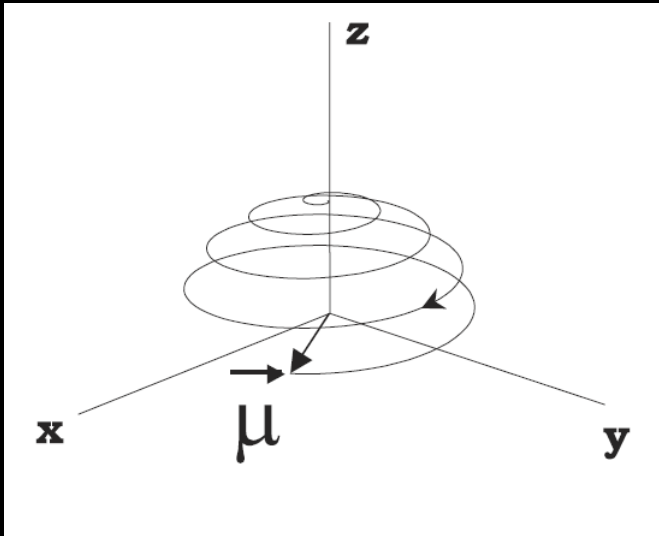
Tipping due to a rotating B_1
Viewed in Laboratory frame

Tipping the Magnetization

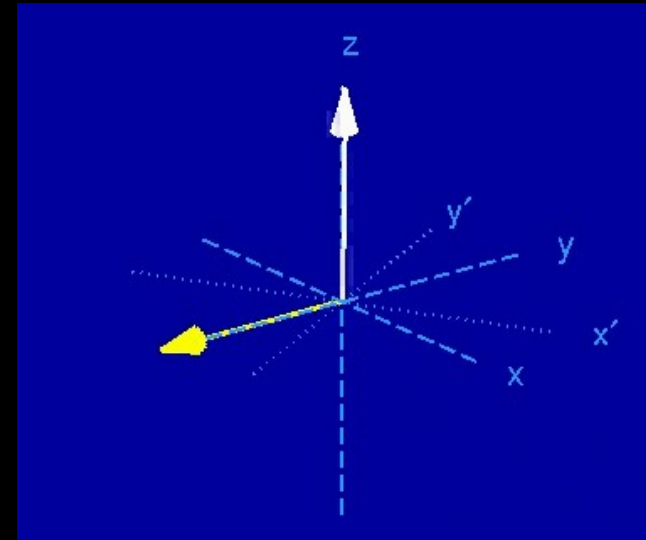


A rotating B_1 field - ON

$$\left(\frac{d\mu}{dt}\right)' = \gamma \cdot \mu \times (B_{1y'})$$

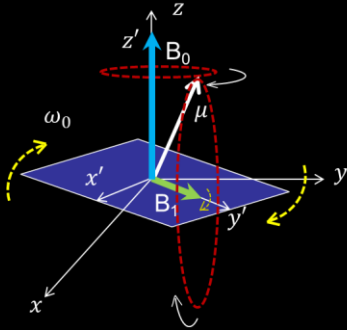


Tipping due to a rotating B_1
Viewed in Laboratory reference
frame



Tipping due to a rotating B_1
Viewed in Rotating reference
frame

Tipping the Magnetization



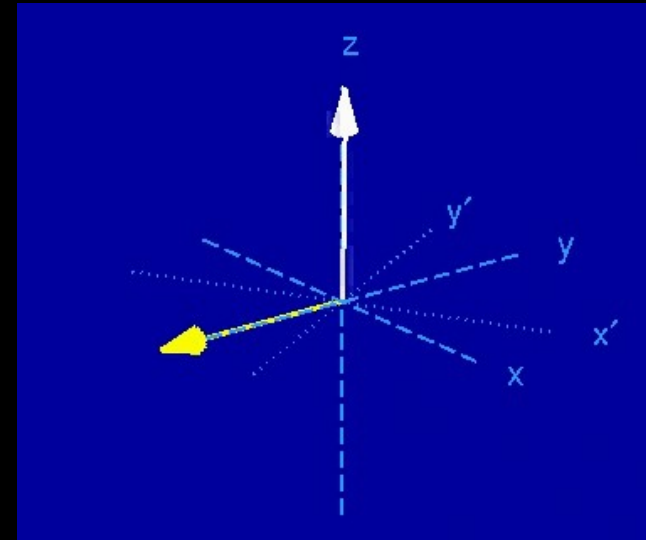
A rotating B₁ field - ON

$$\left(\frac{d\mu}{dt}\right)' = \gamma \cdot \mu \times (B_{1x'})$$

Rate at which the magnetization rotates around B₁ is also given by the Larmor relation:

$$\omega_1 = -\gamma \cdot B_1$$

|B₁| here represents the amplitude of the radiofrequency pulse excitation

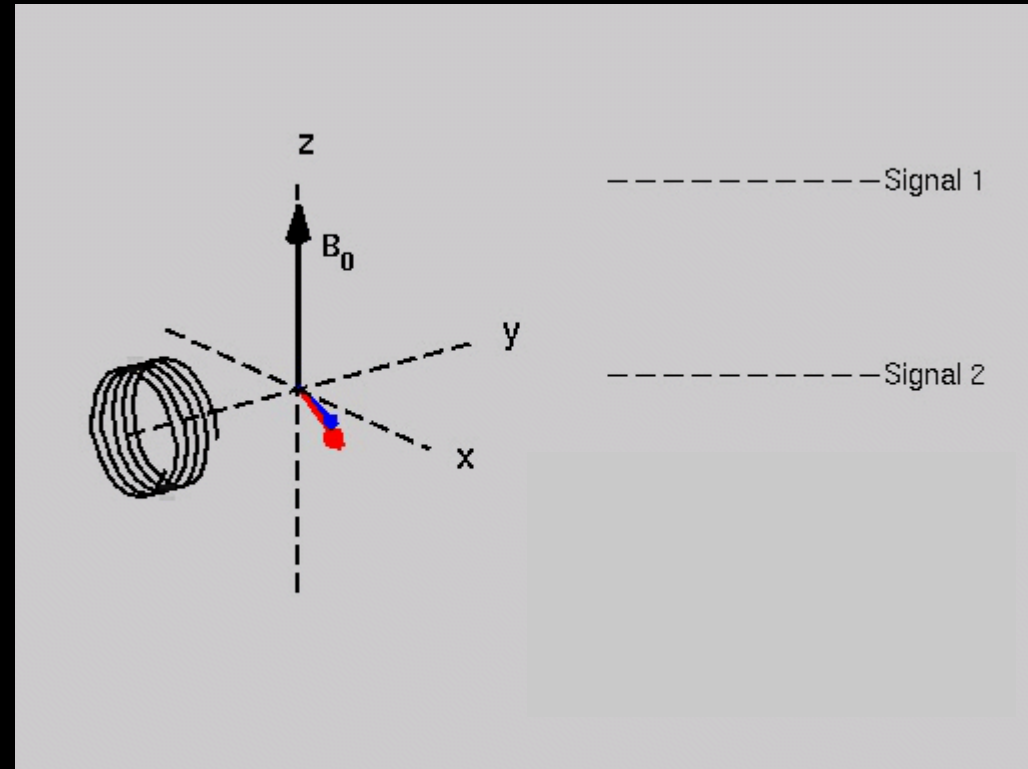
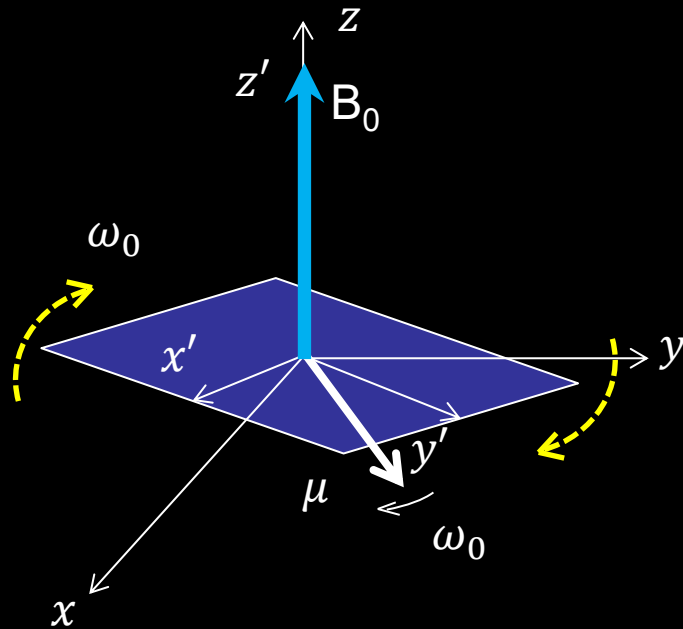


Tipping due to a rotating B₁
Viewed in Rotating reference
frame

Tipping the Magnetization

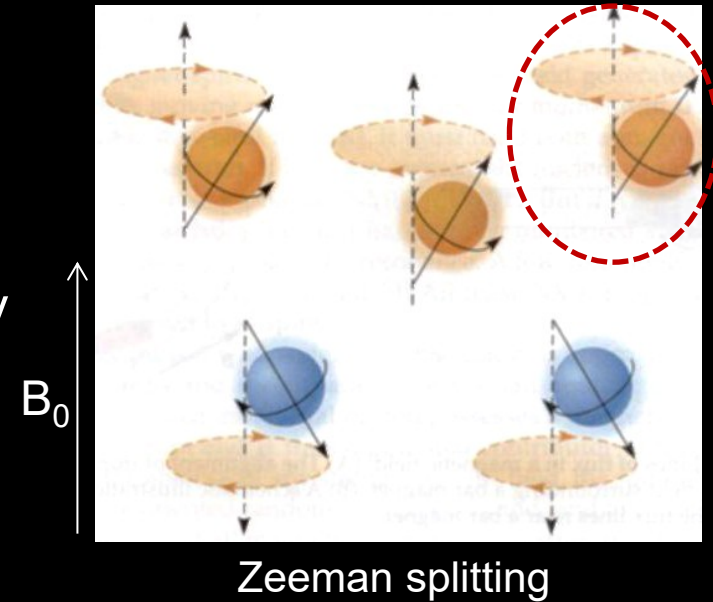
$$\frac{d\mu}{dt} = \gamma \cdot \mu \times B \quad \omega_0 = -\gamma \cdot B_0$$

A rotating B_1 field - OFF



An ensemble of protons/spins

Due to finite thermal energy, some of the spin/magnetization vectors will align parallel (low-energy state) or anti-parallel (high-energy state) to B_0



However, compared to those aligned anti-parallel, a small excess of protons are aligned parallel with the magnetic field

These excess protons are the ones responsible for the measurable NMR/MRI signal

An ensemble of protons/spins

The spin excess

Proportion of the excess spins
among the total available in a
tissue

$$\frac{N(\uparrow)}{N(\downarrow)} = e^{\frac{\Delta E}{k \cdot T}} = p \simeq \frac{\gamma \cdot \hbar \cdot B_0}{2 \cdot k \cdot T}$$

Energy difference here, $\Delta E = \hbar \omega_0$

... i.e. the spin excess is proportional to the imaging
field strength

...this is the primary motivation
for going to higher imaging
field strengths

An ensemble of protons/spins

The spin excess

For hydrogen protons in our body,

$$p = 3 \times 10^{-6},$$

i.e. in each 1 million spins, there are 3 spins parallel to B_0 in excess of those which are anti-parallel.

In Hyper polarized MRI for Lung imaging, noble gases like ^3He can be polarized to an $p = 0.6$, a 10^5 fold increase.



NMR – Proton (H^1) and others

Although we are mainly concerned with Hydrogen proton MRI here, in general any nucleus with a non-zero magnetic moment and spin can be investigated using NMR

... typically those with odd atomic mass and atomic number (i.e. unequal number of neutrons and protons in their nuclei)

^{23}Na , ^{31}P , ^{17}O , ^{19}F , are some of the other biologically relevant nuclei that are conducive to NMR

Nucleus i	γ_i (MHz/T)
^1H , gray matter	42.58
^{23}Na , average tissue	11.27
^{31}P , average tissue	17.25
^{17}O , gray matter	−5.77
^{19}F , average tissue	40.08

Self-study recommendation

Chapter 2	Problem #s - 2.1, 2.2, 2.4, 2.6, 2.8
Chapter 3	Problem #s 3.3, 3.6

MR signal detection and Image formation

Magnetization Notation

$$p \simeq \frac{\gamma \cdot \hbar \cdot B_0}{2 \cdot k \cdot T}$$

μ

→

M_0

Magnetization - Single proton

Net Magnetization - Proton ensemble
(spin excess)

Magnitude of M_0 (for an arbitrary nucleus with spin quantum number s),

$$M_0 \propto \rho_0 \frac{s \cdot (s + 1) \cdot \gamma^2 \cdot \hbar^2 \cdot B_0}{3 \cdot k \cdot T}$$

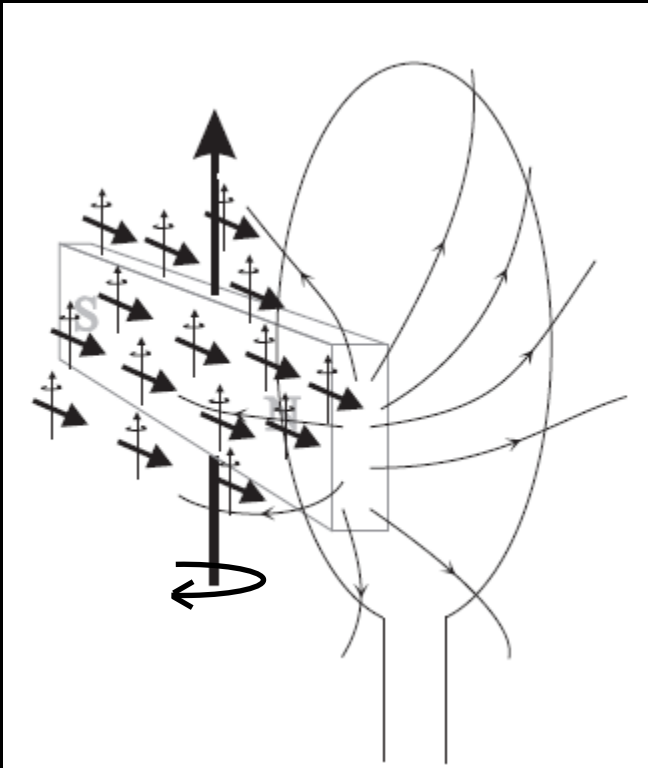
Magnitude of M_0 (for ^1H ,
 $S=1/2$),

$$M_0 \simeq \rho_0 \frac{\gamma^2 \cdot \hbar^2 \cdot B_0}{4 \cdot k \cdot T}$$

ρ_0 is the # of spins per unit
volume

Signal Detection

Faraday's Law of Electromagnetic induction



$$emf = -\frac{d\Phi}{dt}$$

Induced signal is proportional to rate of change of flux

$$\omega_0 = -\gamma \cdot B_0$$

$$\Rightarrow emf \propto \omega_0$$

$$signal \propto \omega_0 \cdot M_0 \cdot V_s \cdot e^{i(\omega_0 t + \phi)}$$

V_s is voxel volume

Signal Detection

$$\text{signal: } s(t) \propto \gamma \cdot B_0 \cdot M_0 \cdot V_s \cdot e^{i(\omega_0 t + \phi)}$$

V_s is voxel volume

$$M_0 \simeq \rho_0 \frac{\gamma^2 \cdot \hbar^2 \cdot B_0}{4 \cdot k \cdot T}$$

$$\Rightarrow \text{signal} \propto B_0^2$$

... one more motivation for
higher field strength imaging

Encoding spatial information: Imaging

The goal is to differentiate signals coming from different spatial locations

In the early 1970s both Paul Lauterbur and Peter Mansfield realized that by forcing the B_0 field to vary on purpose, one could recover such spatial information from the object

Imaging

Their approach was that by applying a magnetic field gradient in one direction one could spatially distinguish spins along that direction in terms of their precession frequency, i.e., if

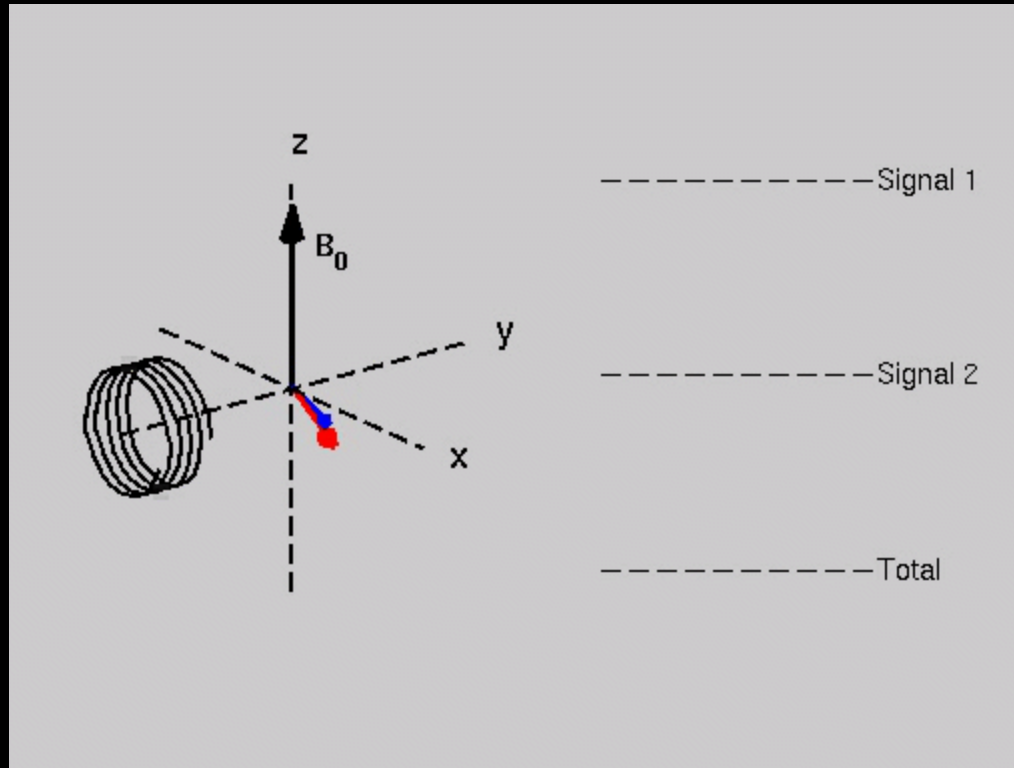
$$B(x) = B_0 + G_x \cdot x$$

$$\omega(x) = \omega_0 + \gamma \cdot G_x \cdot x$$

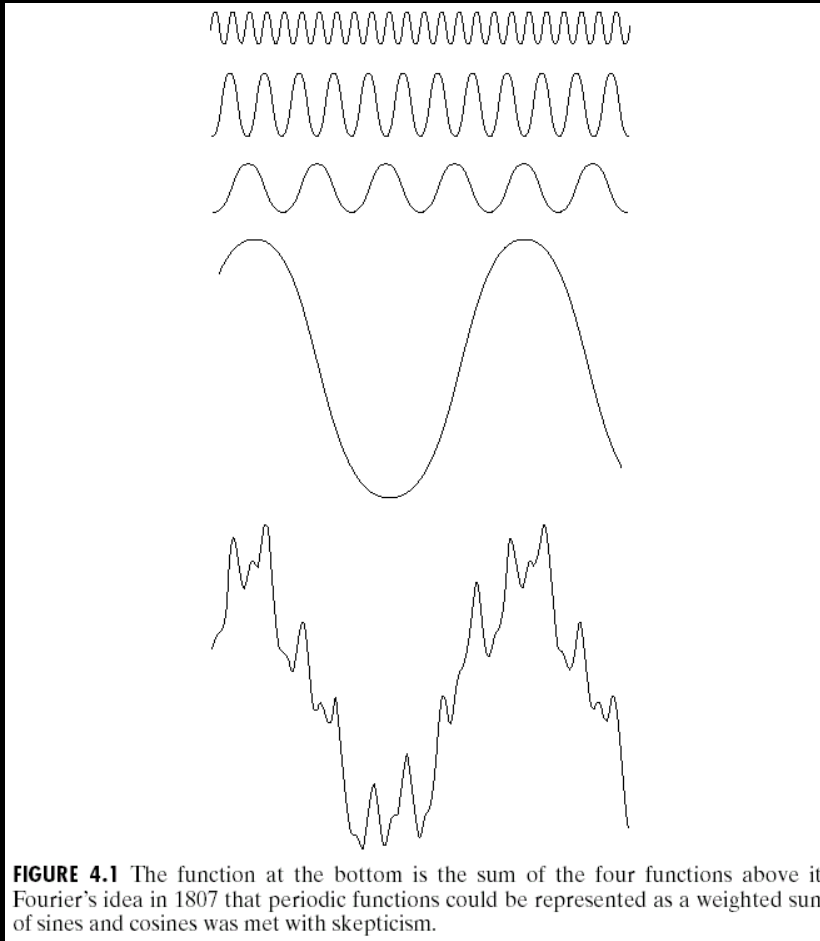
... i.e., for every position x in the object there is a unique frequency $\omega(x)$ whose spectral (signal) amplitude is related to the number of spins per unit volume at that position.

Signal Reception

The signal we measure:



What is a Fourier Transform?



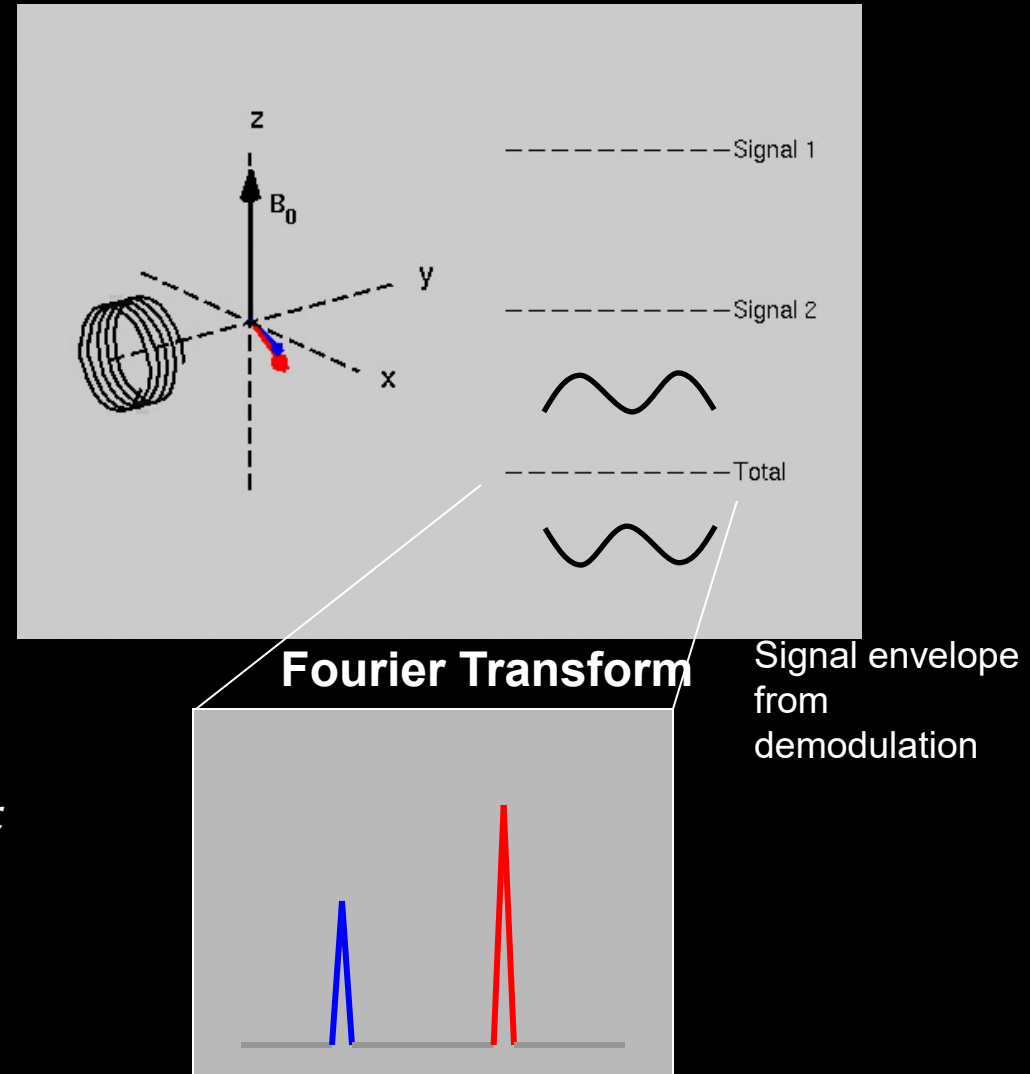
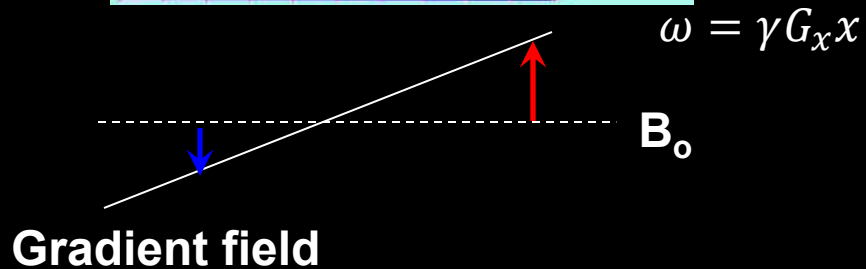
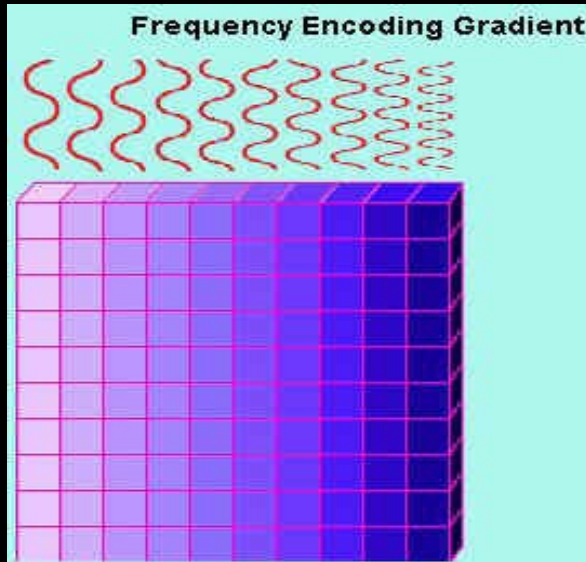
Provides us with a mathematical framework to express any signal, within a specific width of time/space, be expressed as sum of complex sines and cosines.

$$\mathfrak{F}\{f(x)\} = F(u) = \int_{-\infty}^{\infty} f(x) \exp[-j2\pi ux] dx$$

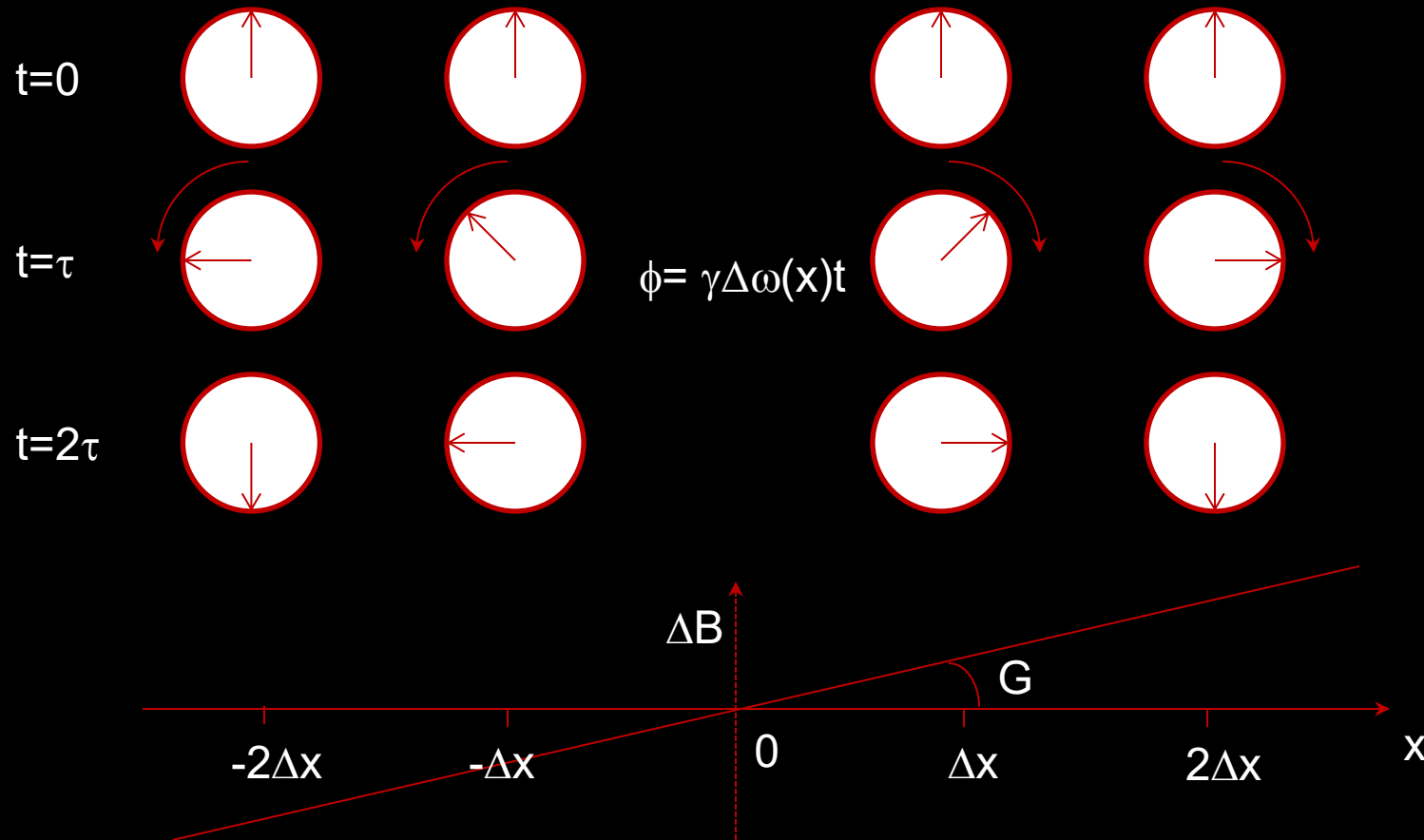
$$\mathfrak{F}^{-1}\{F(u)\} = f(x) = \int_{-\infty}^{\infty} F(u) \exp[j2\pi ux] du$$

Brief introduction: Spatial Encoding

Encoding Position through
Magnetic field gradient: 1D
example



Frequency encoding



Frequency encoding

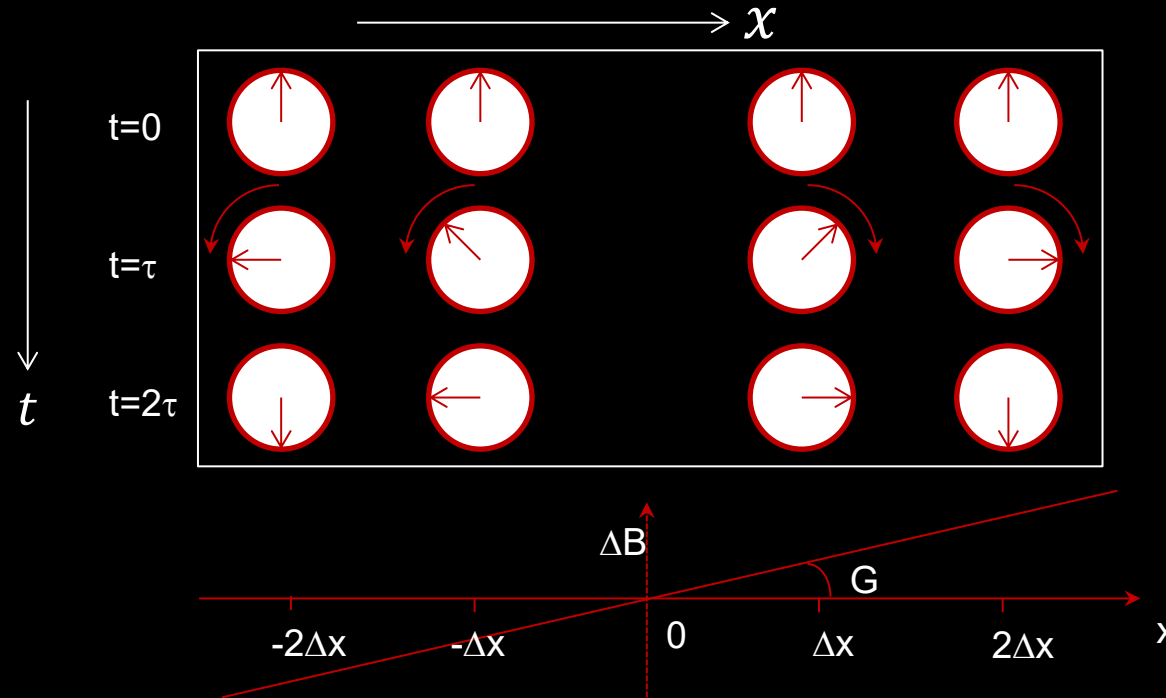
$$\varphi(x) = \omega(x) \cdot t$$

$$\varphi(x) = -\gamma \cdot (B_0 + G_x \cdot x) \cdot t$$

Rotating reference frame –
signal demodulation

$$\Delta\varphi(x) = -\gamma \cdot G_x \cdot x \cdot t = -2 \cdot \pi \cdot k_x \cdot x$$

Frequency encoding



$$s(t) = \int dx \cdot \rho(x) \cdot e^{i \cdot \Delta \varphi(x,t)} \longrightarrow s(t) = \int dx \cdot \rho(x) \cdot e^{-i \cdot \gamma \cdot G_x \cdot x \cdot t}$$

$s(t)$ contains signal contribution from all excited spins in the sample

1D Fourier Transform

$$\Delta\varphi(x) = -\gamma \cdot G_x \cdot x \cdot t = -2 \cdot \pi \cdot k_x \cdot x$$

$$s(k_x) = \int dx \cdot \rho(x) \cdot e^{-i \cdot 2 \cdot \pi \cdot k \cdot x} \longrightarrow FT[\rho(x)]$$

$$\begin{aligned} \rho(x) &= IFT[s(k)] \\ &= \int dk \cdot s(k_x) \cdot e^{i \cdot 2 \cdot \pi \cdot k \cdot x} \end{aligned}$$

... thus we are able to reconstruct the 1D profile of the spin density along X direction using a linear magnetic field gradient along X.

The spatial frequency variable k

$$\Delta\varphi(x) = -\gamma \cdot G_x \cdot x \cdot t = -2 \cdot \pi \cdot k_x \cdot x$$

$$k_x = \frac{\gamma}{2\pi} \cdot G_x \cdot t = \gamma \cdot G_x \cdot t$$

$$k_x = \gamma \int dt \cdot G_x(t)$$

Similar to ω , which represents time-frequency, k represents spatial frequency.

k has the units of per meter

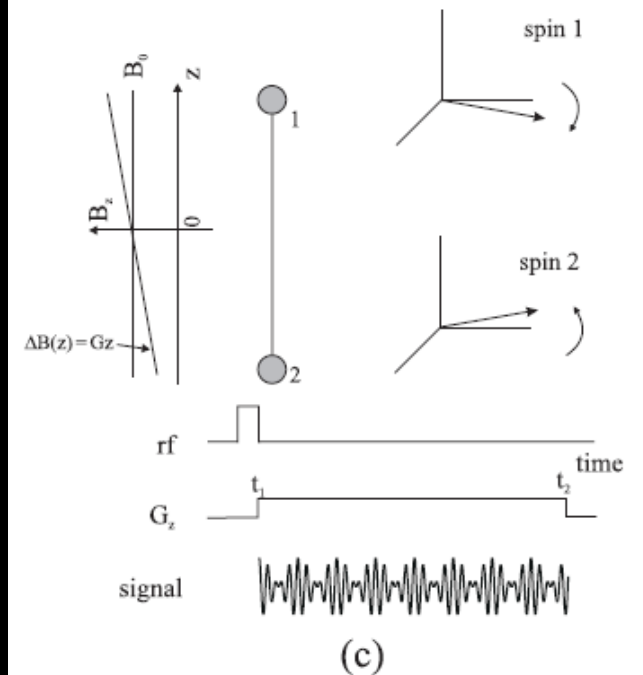
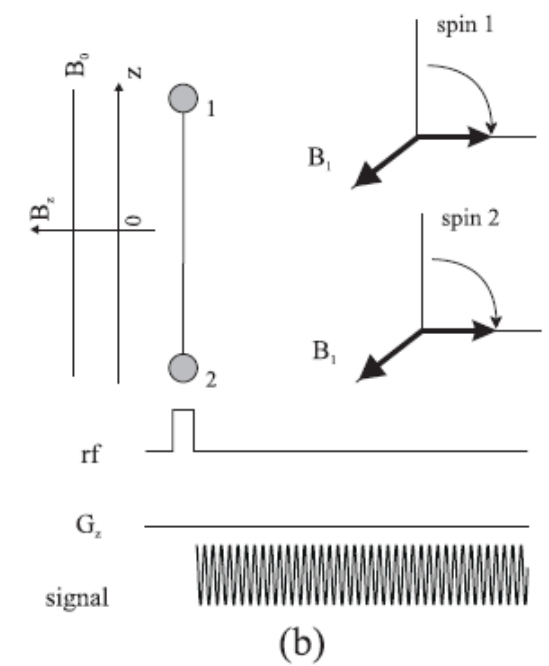
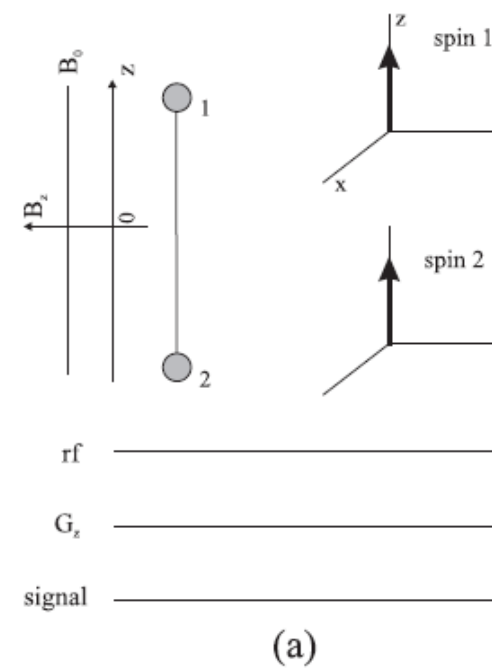
Two spin example analysis

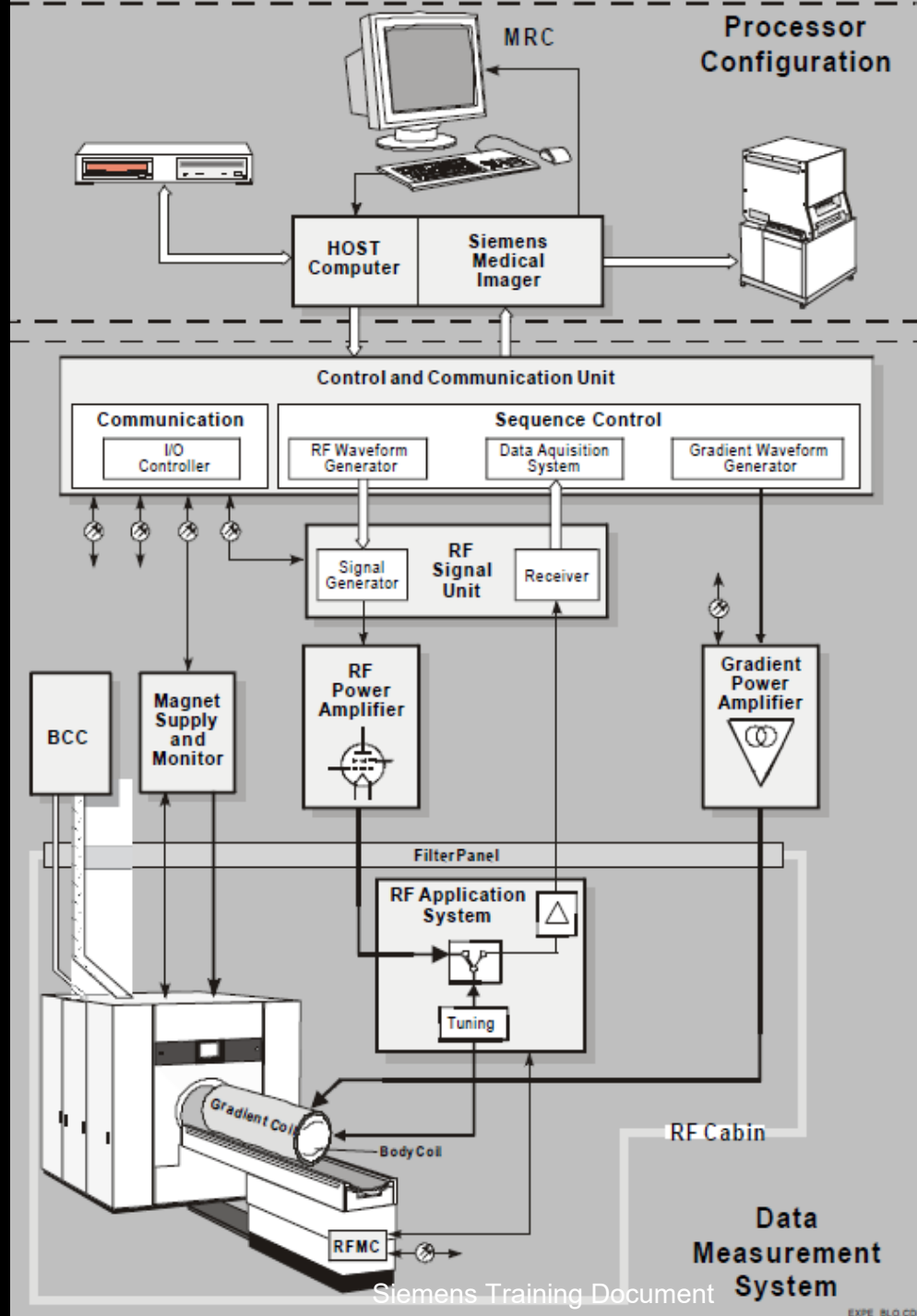
$$s(t) = S_0(e^{-i\gamma G z_0 t} + e^{i\gamma G z_0 t}) \\ = 2S_0 \cos(\gamma G z_0 t)$$

$$s(k) = 2S_0 \cos(2\pi k z_0)$$



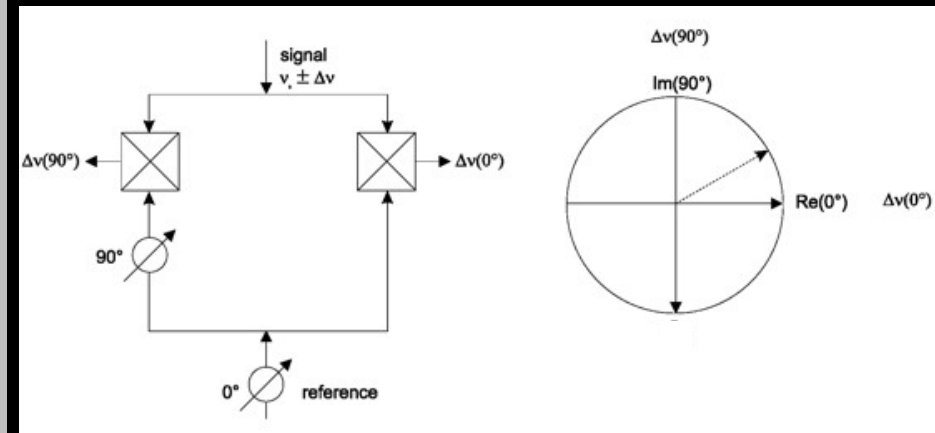
$$\rho(z) = \int_{-\infty}^{\infty} dk 2S_0 \cos(2\pi k z_0) e^{i2\pi k z} \\ = S_0 \int_{-\infty}^{\infty} dk (e^{i2\pi k(z+z_0)} + e^{i2\pi k(z-z_0)}) \\ = S_0 [\delta(z + z_0) + \delta(z - z_0)]$$



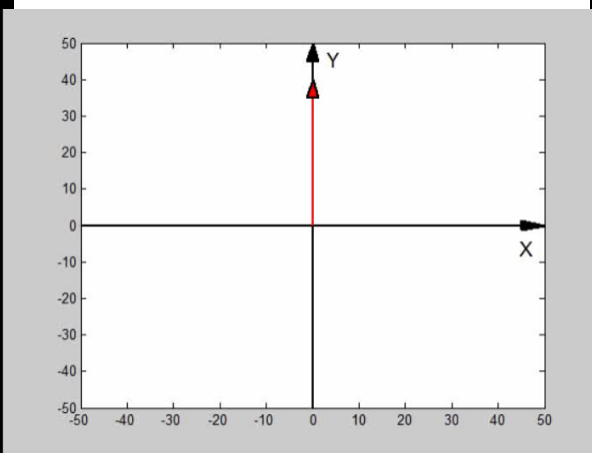
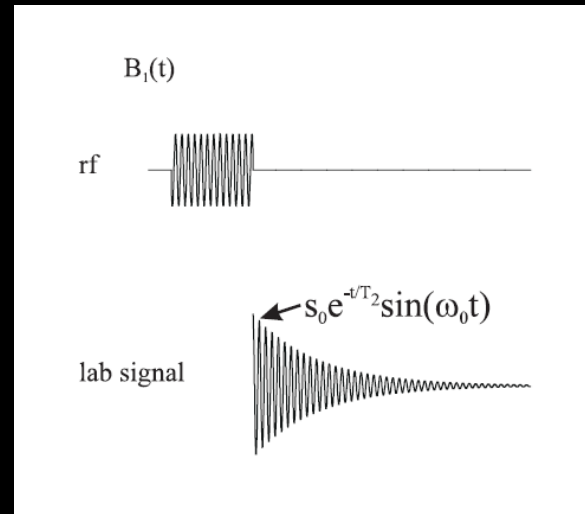


Signal acquisition

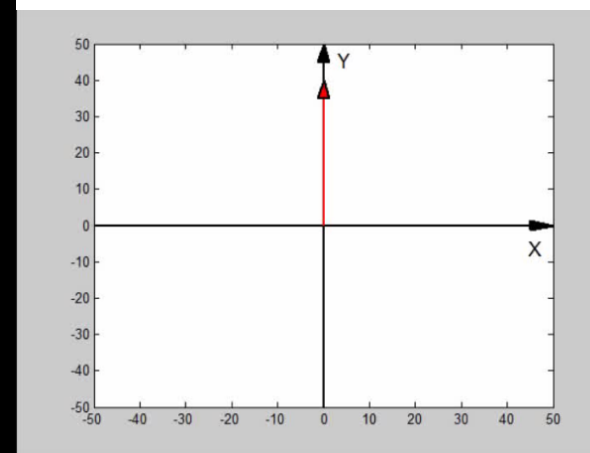
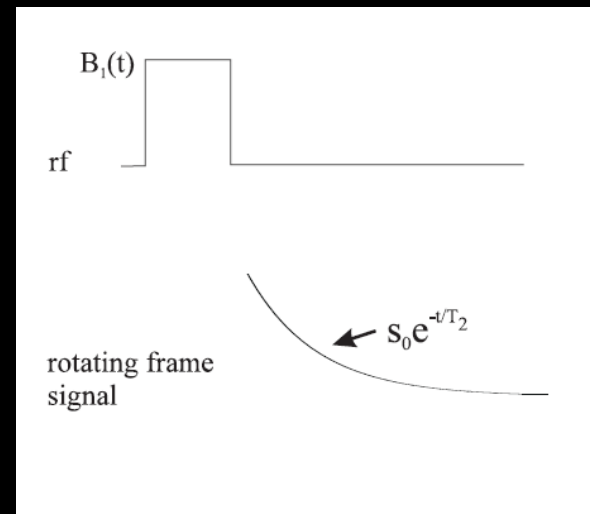
Quadrature demodulation:



Free induction decay



Demodulated signal

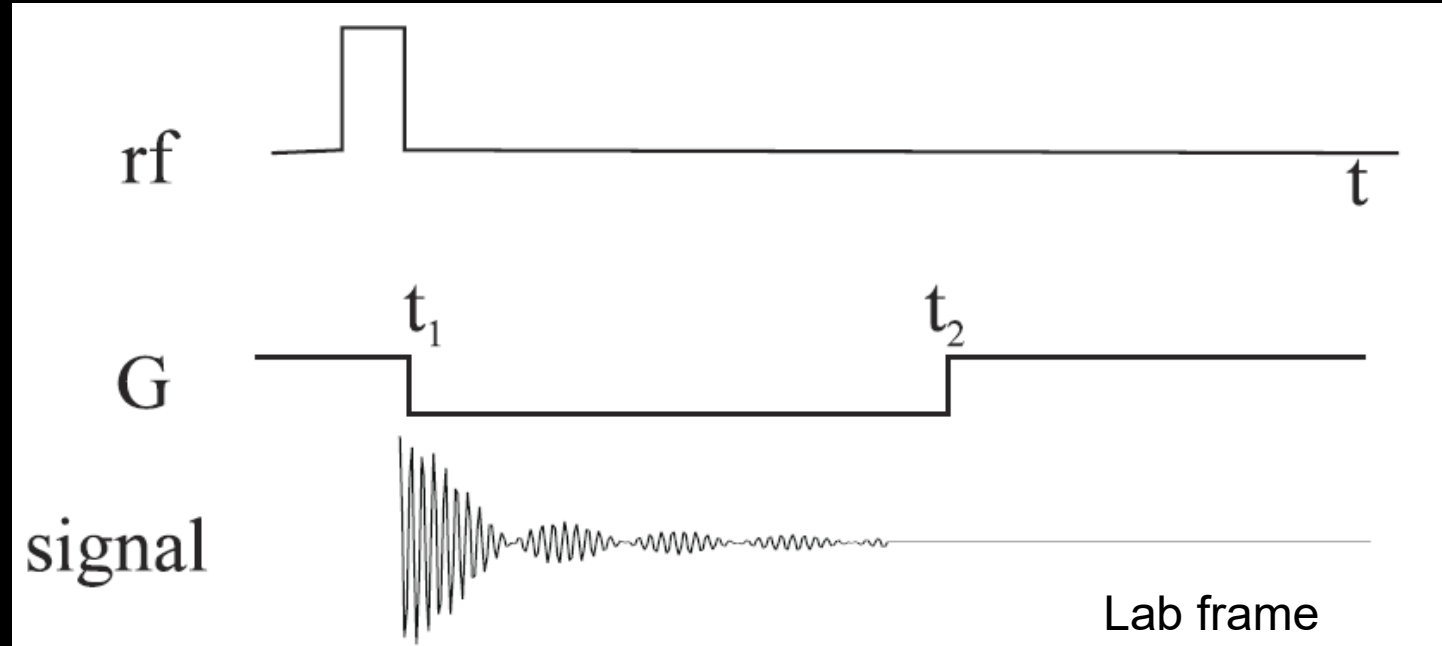


$$e^{-\frac{t}{T_2}}$$

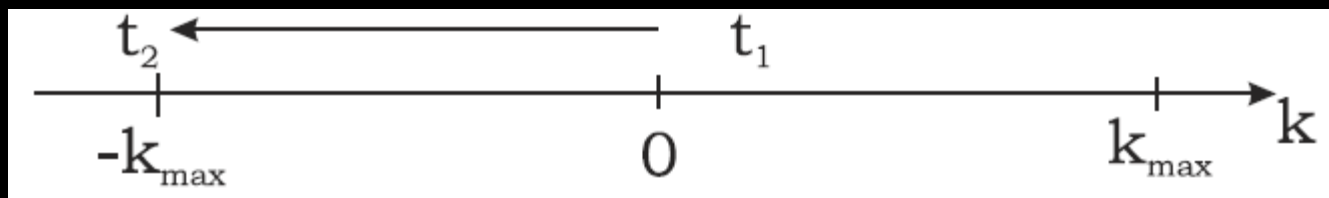
Signal decay occurs due to tiny differences in the precession frequency of the spins that lead them to get out of phase/sync – characterized by time T_2

Free induction decay

Gradient ON

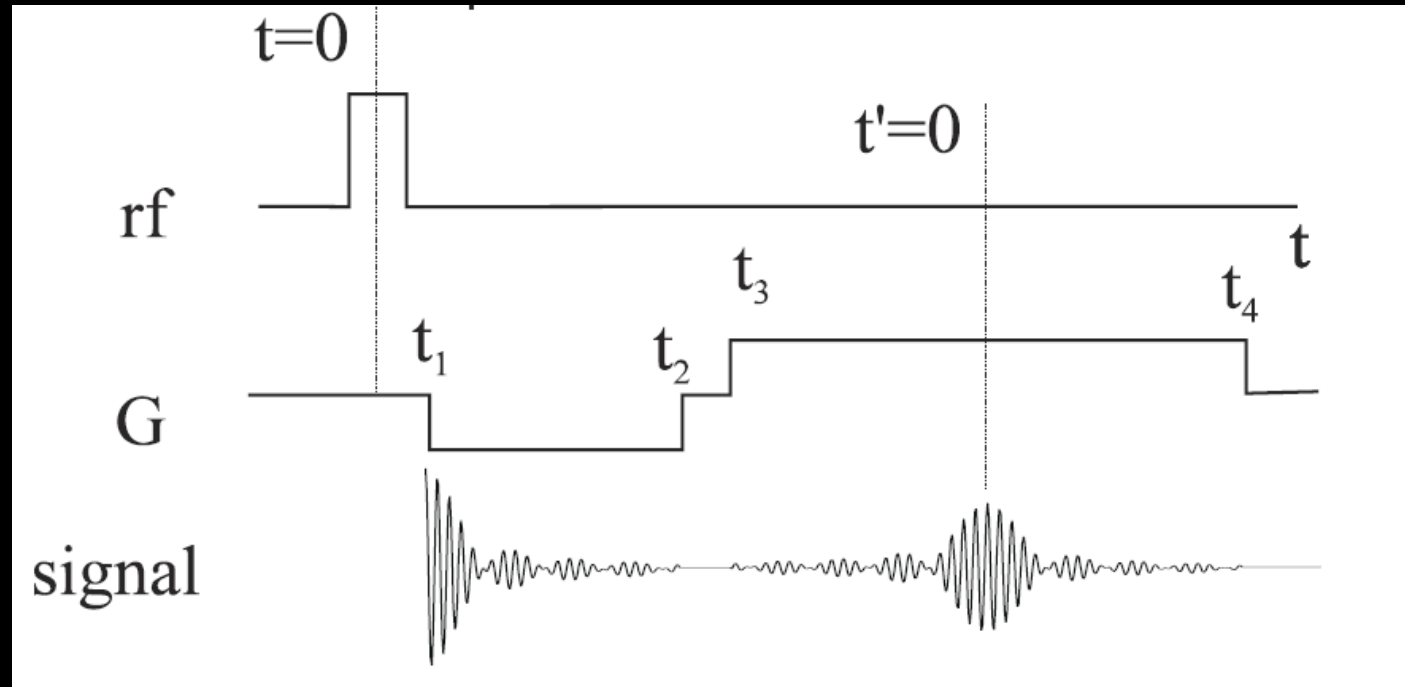


$$k_x = \gamma \cdot G_x \cdot t$$

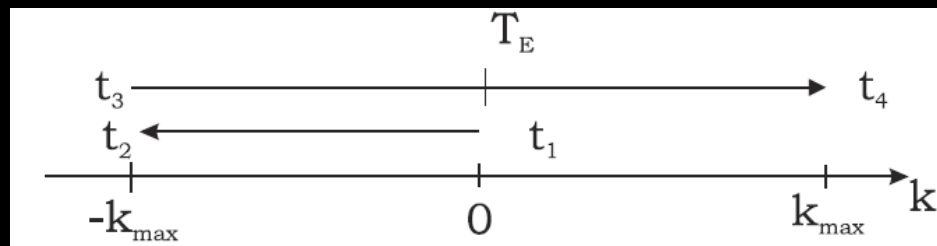


Free induction decay

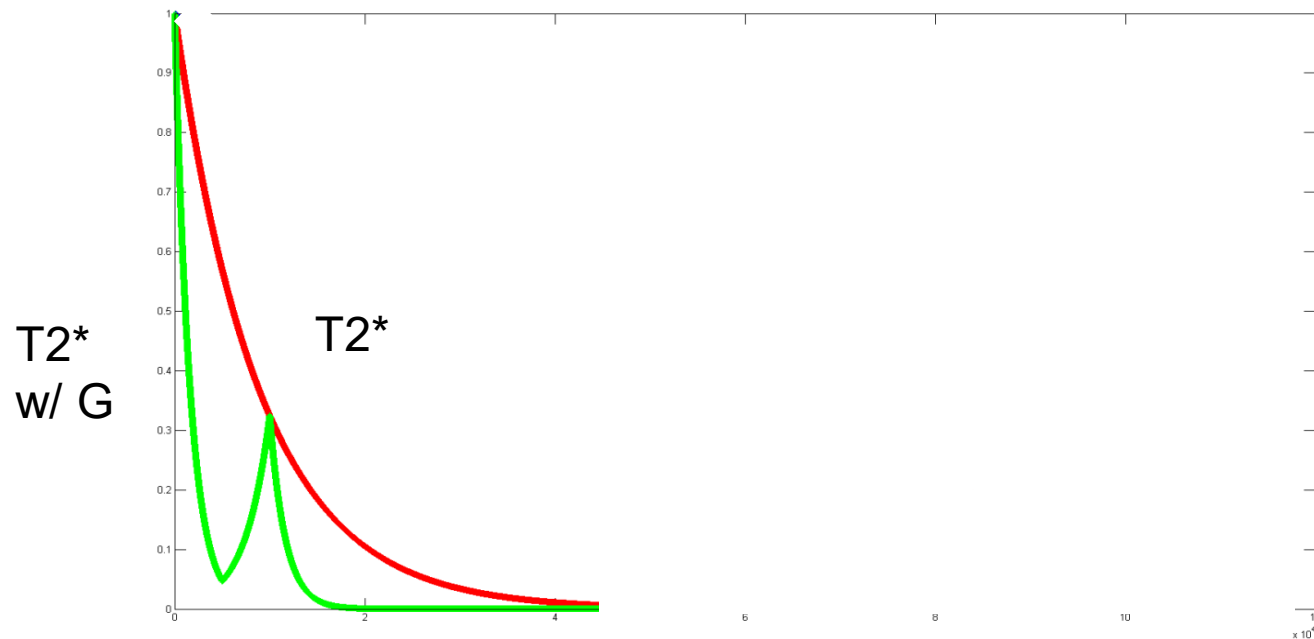
Gradient Echo



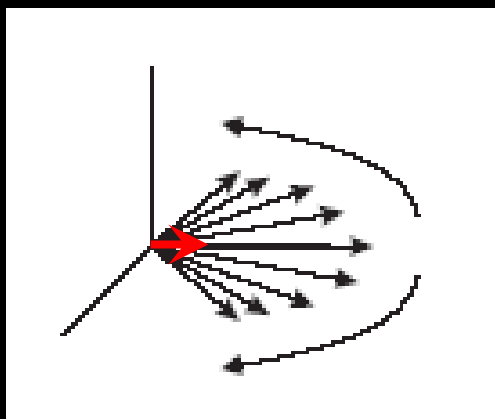
$$k_x = \gamma \cdot G_x \cdot t$$



Relative signal decay



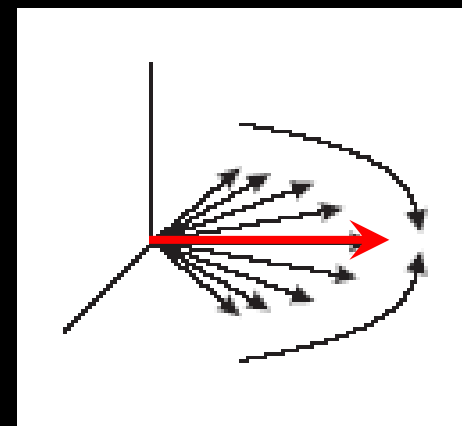
Gradient echo ...



B_0

$-G_x x$

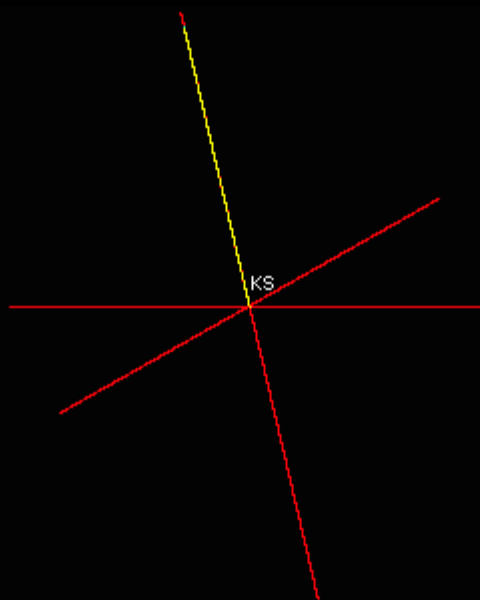
Dephase



B_0

$+G_x x$

Rephase



Gradient Echo

Excitation pulse



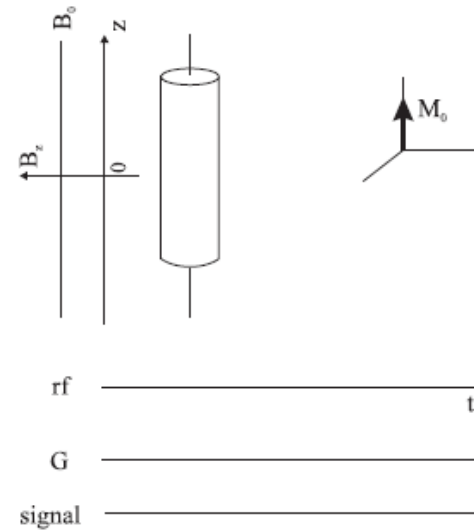
De-phase gradient



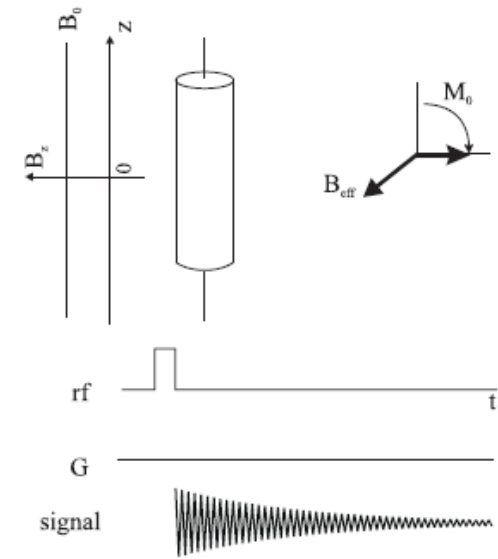
Read gradient

Rule of thumb:

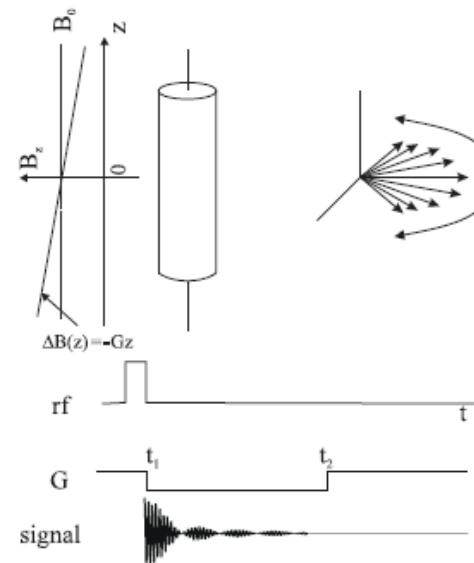
- Net Read gradient moment (i.e. $\int Gt$) is zero at the echo center



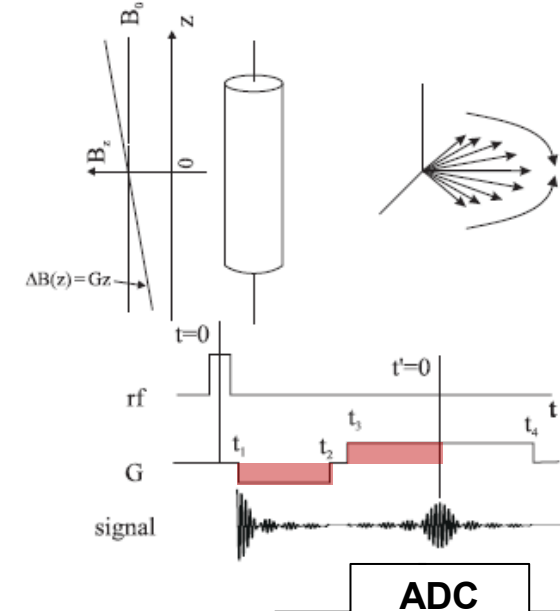
(a)



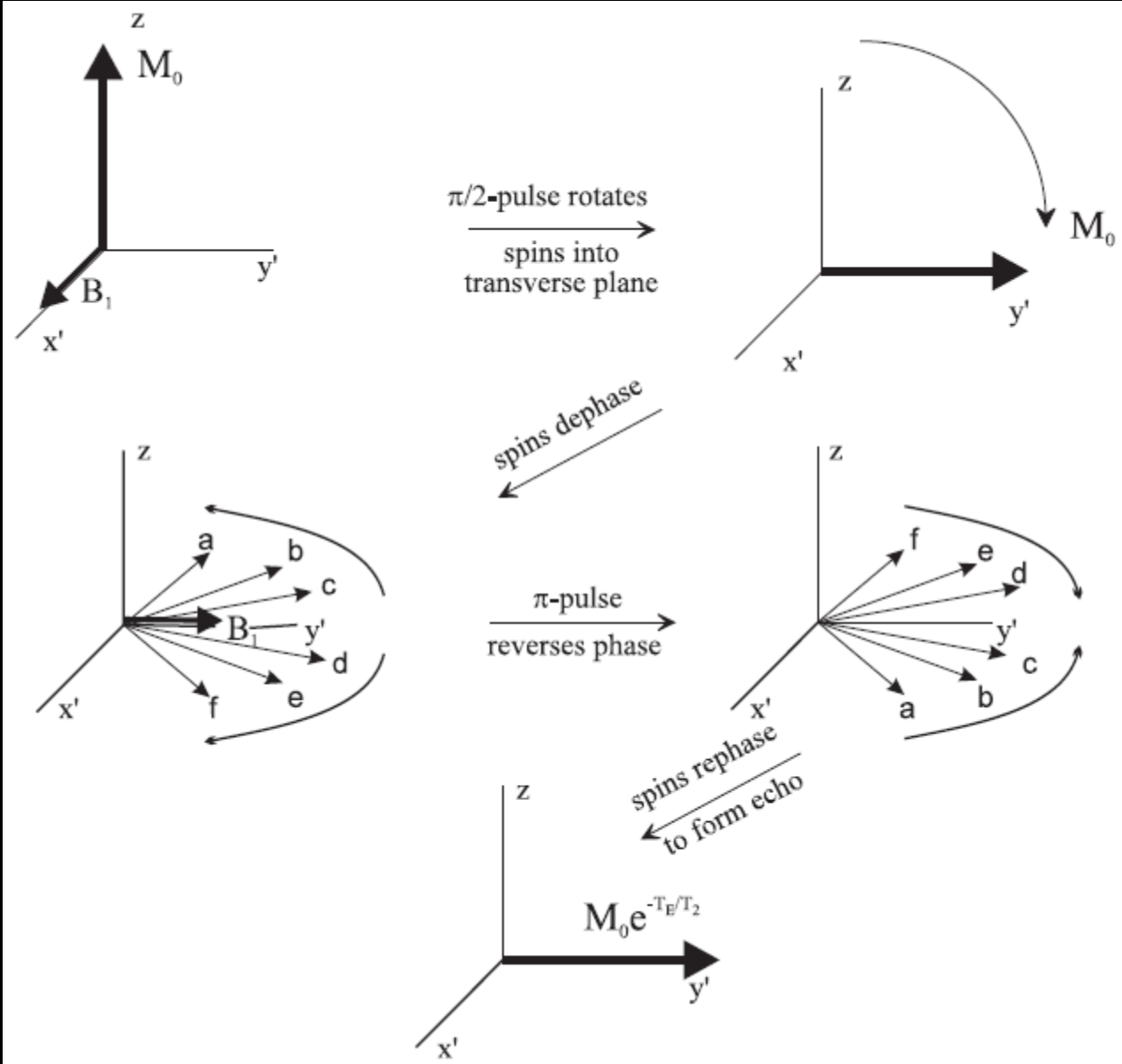
(b)



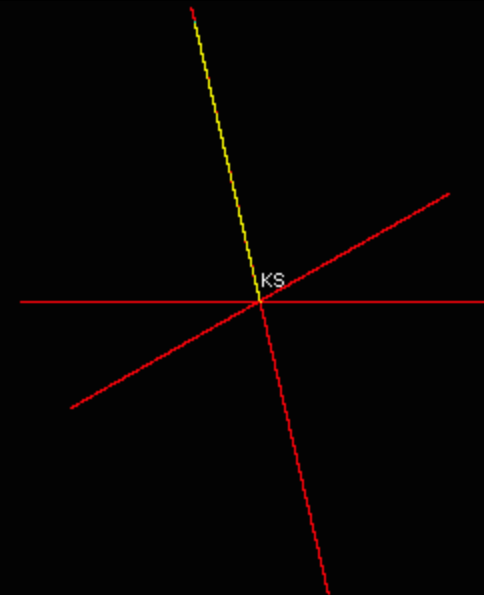
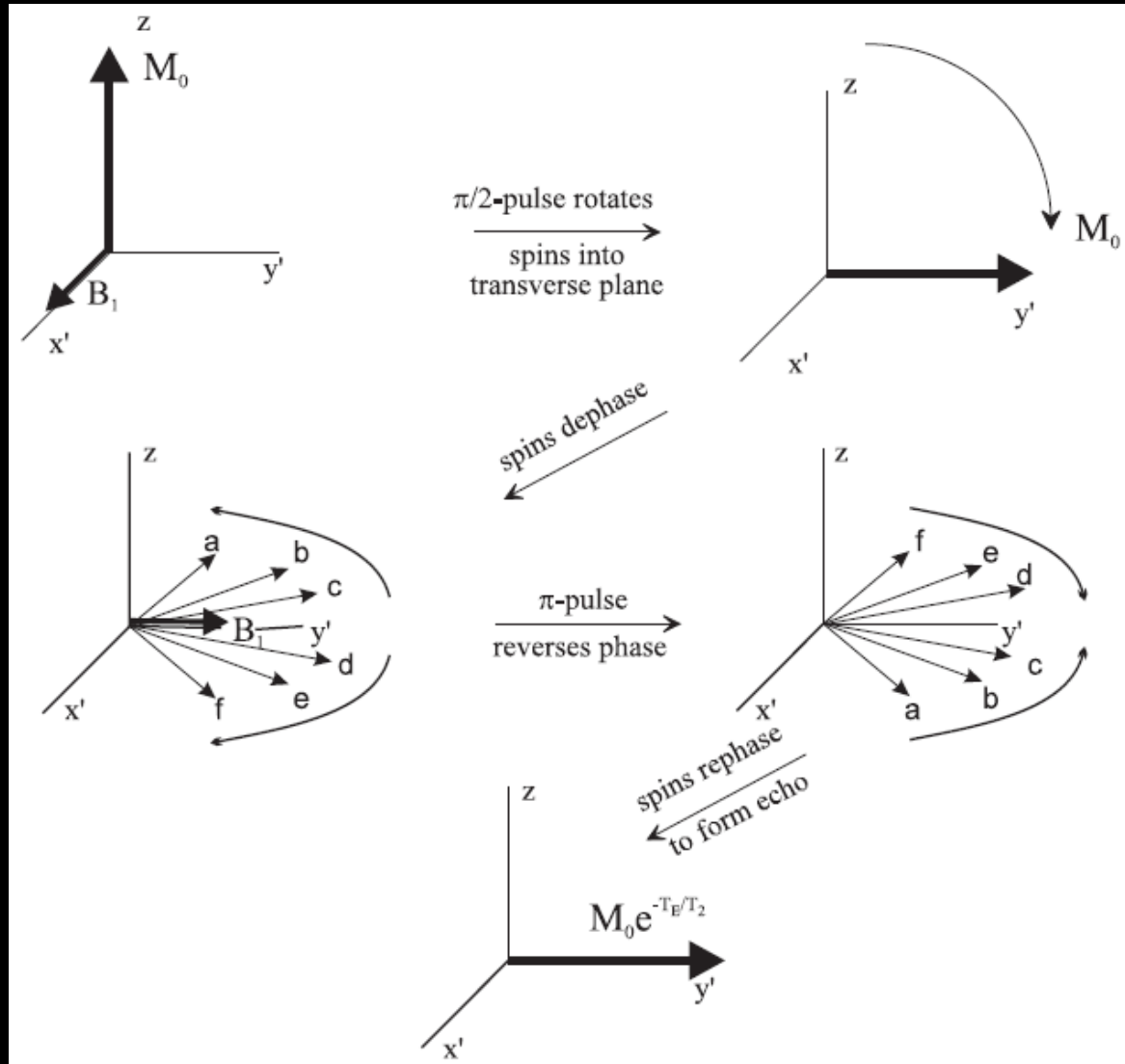
(c)



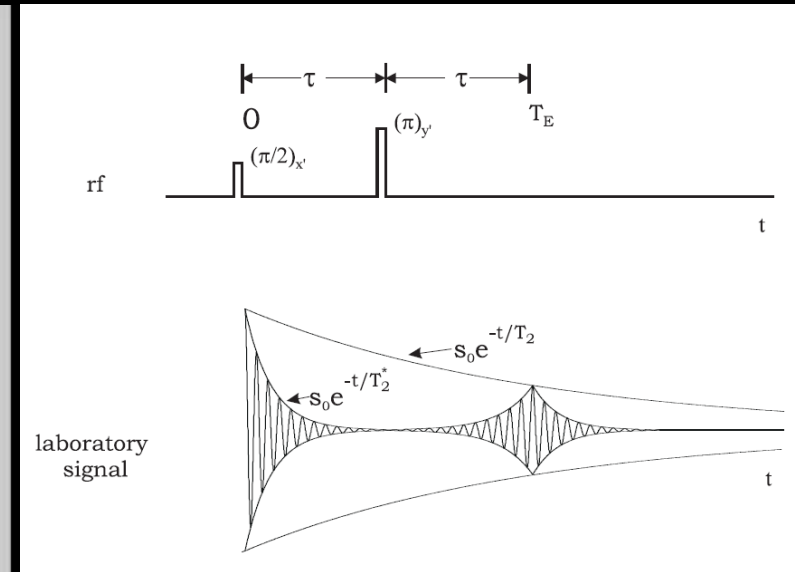
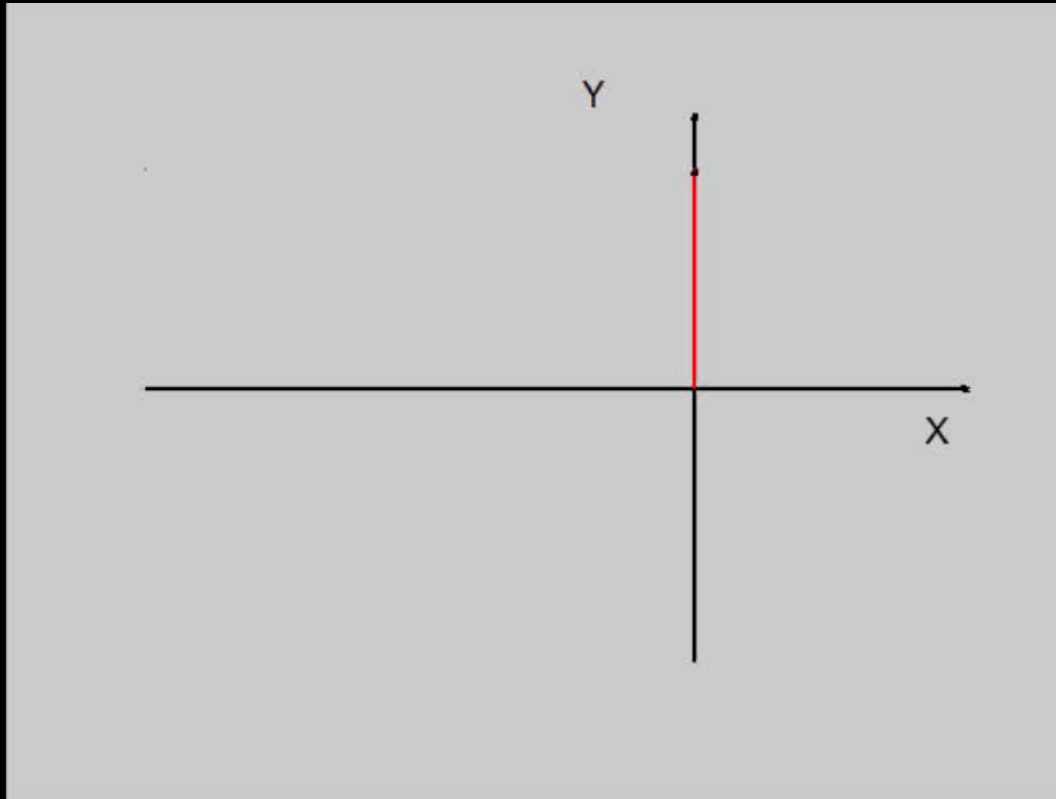
Spin echo (rf echo)



Spin echo (rf echo)



Spin echo envelope

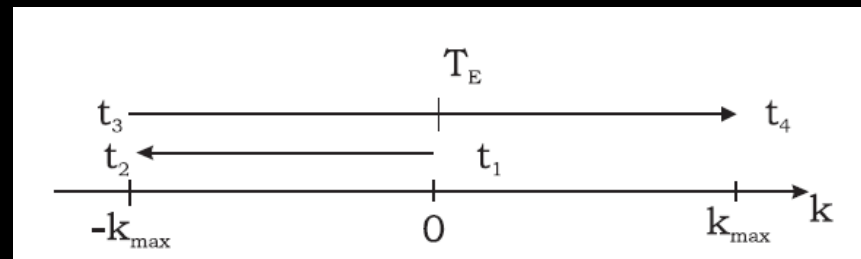
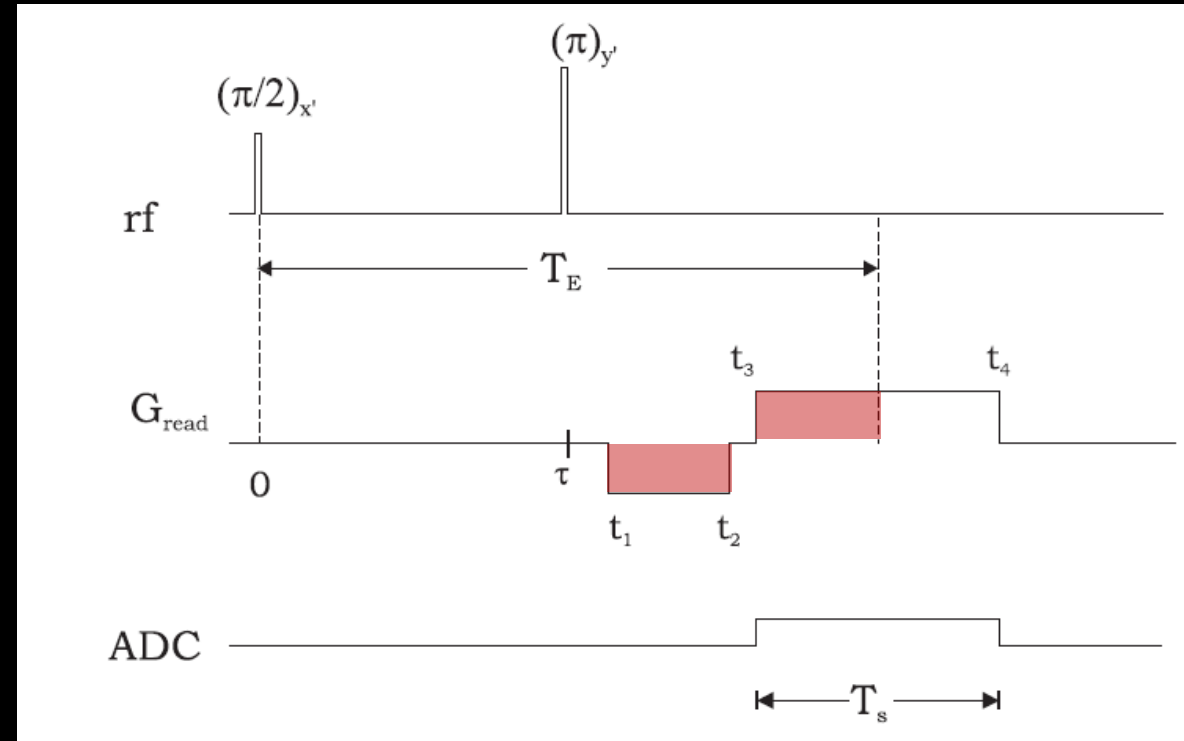


You do not need a gradient to get a spin echo

... however, you need a gradient to encode spatial information (i.e. to generate an image)

Spin echo

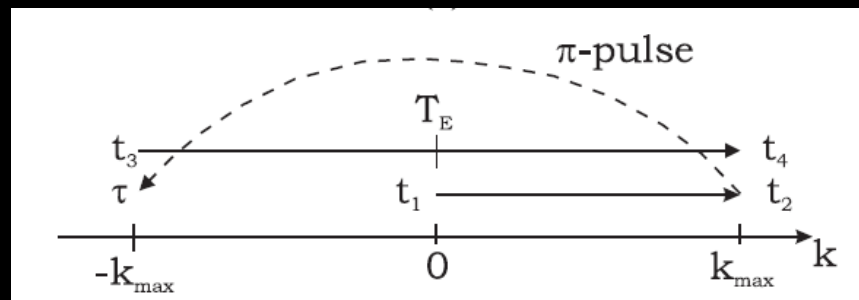
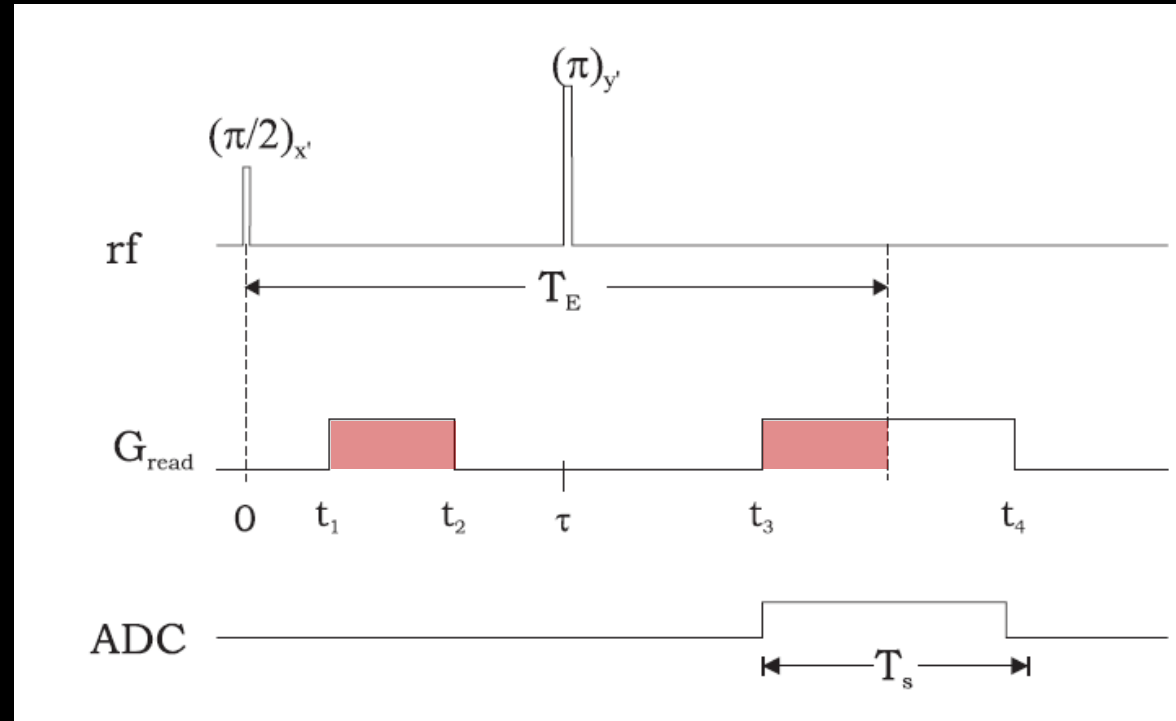
W/Spatial encoding



Spin echo

W/Spatial encoding

...alternatively



Multidimensional Spatial Encoding (2D/3D imaging)

1D: $s(k_x) = \int dx \cdot \rho(x) \cdot e^{-i \cdot 2 \cdot \pi \cdot k \cdot x}$



$$x \rightarrow \vec{r} = (x, y, z)$$

3D: $s(k) = \int d^3r \cdot \rho(r) \cdot e^{-i2\pi(\vec{r} \cdot \vec{k})}$

- 1) All dimensions are equivalent
- 2) k along each direction is determined by the gradient in that direction

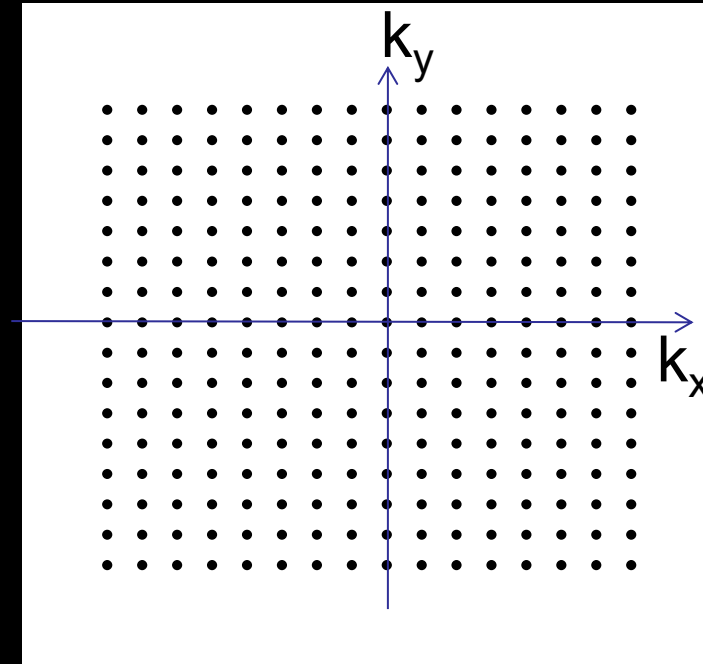
$$\begin{aligned} & s(k_x, k_y, k_z) \\ &= \iiint dx \cdot dy \cdot dz \cdot \rho(x, y, z) \cdot e^{-i2\pi(k_x x + k_y y + k_z z)} \end{aligned}$$

Multidimensional Spatial Encoding (2D/3D imaging)

Practically

- The goal:
Collect every point in the 3 dimensional k-space
- The means
Use gradients to traverse the k-space

$$k_i(t) = \gamma \cdot G_i \cdot i \cdot t, i = x, y, z$$

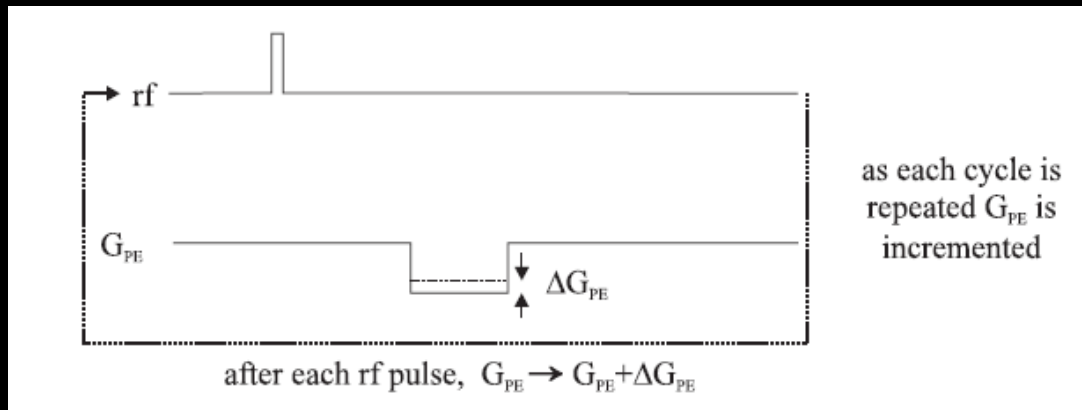
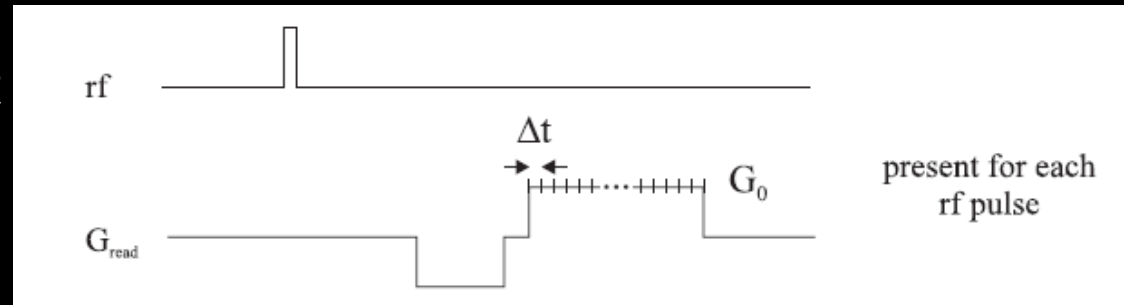


Phase encoding

Frequency encoding

$$k_x = \gamma \cdot G_x \cdot t = \gamma \cdot G_x \cdot \Delta t$$

Encoding phase is accumulated with the progression of time



as each cycle is repeated G_{PE} is incremented

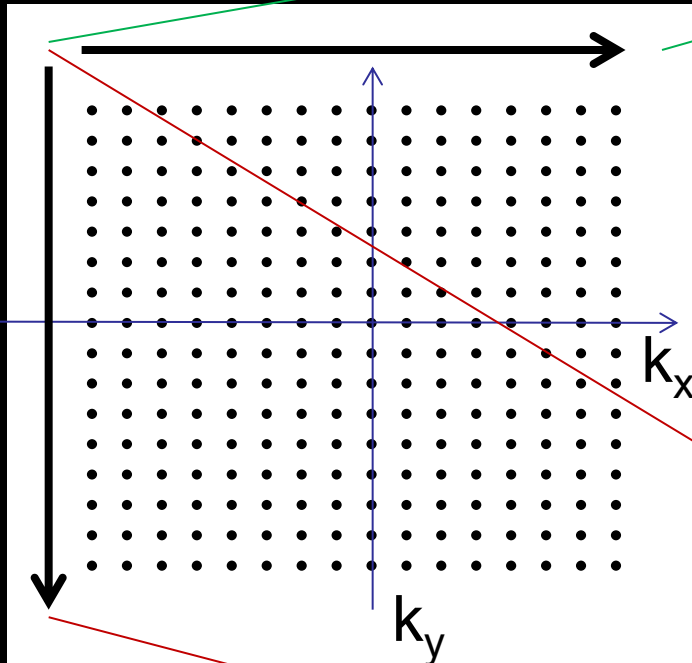
$$k_y = \gamma \cdot G_y \cdot t$$

$$= \gamma \cdot \Delta G_y \cdot \tau$$

Encoding phase is accumulated in steps of the phase encoding gradient

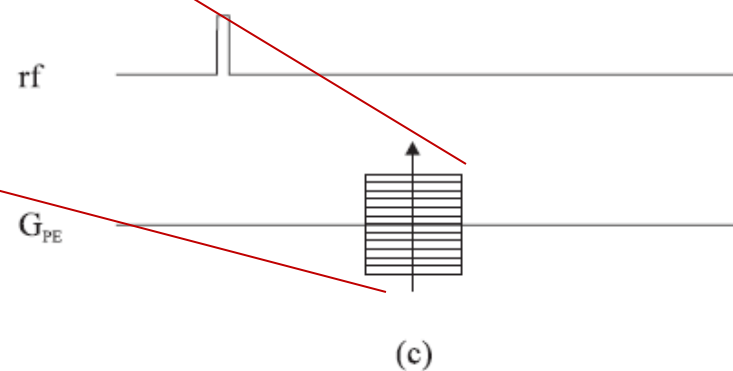
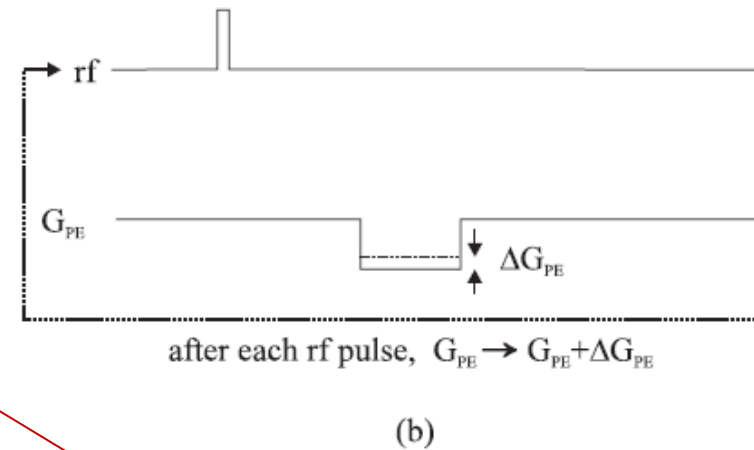
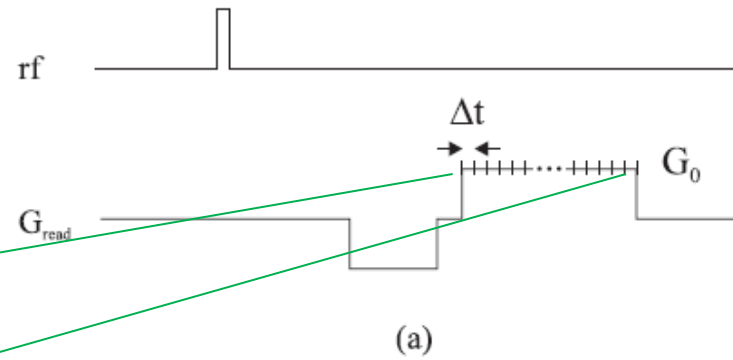
Phase encoding

Phase encoding

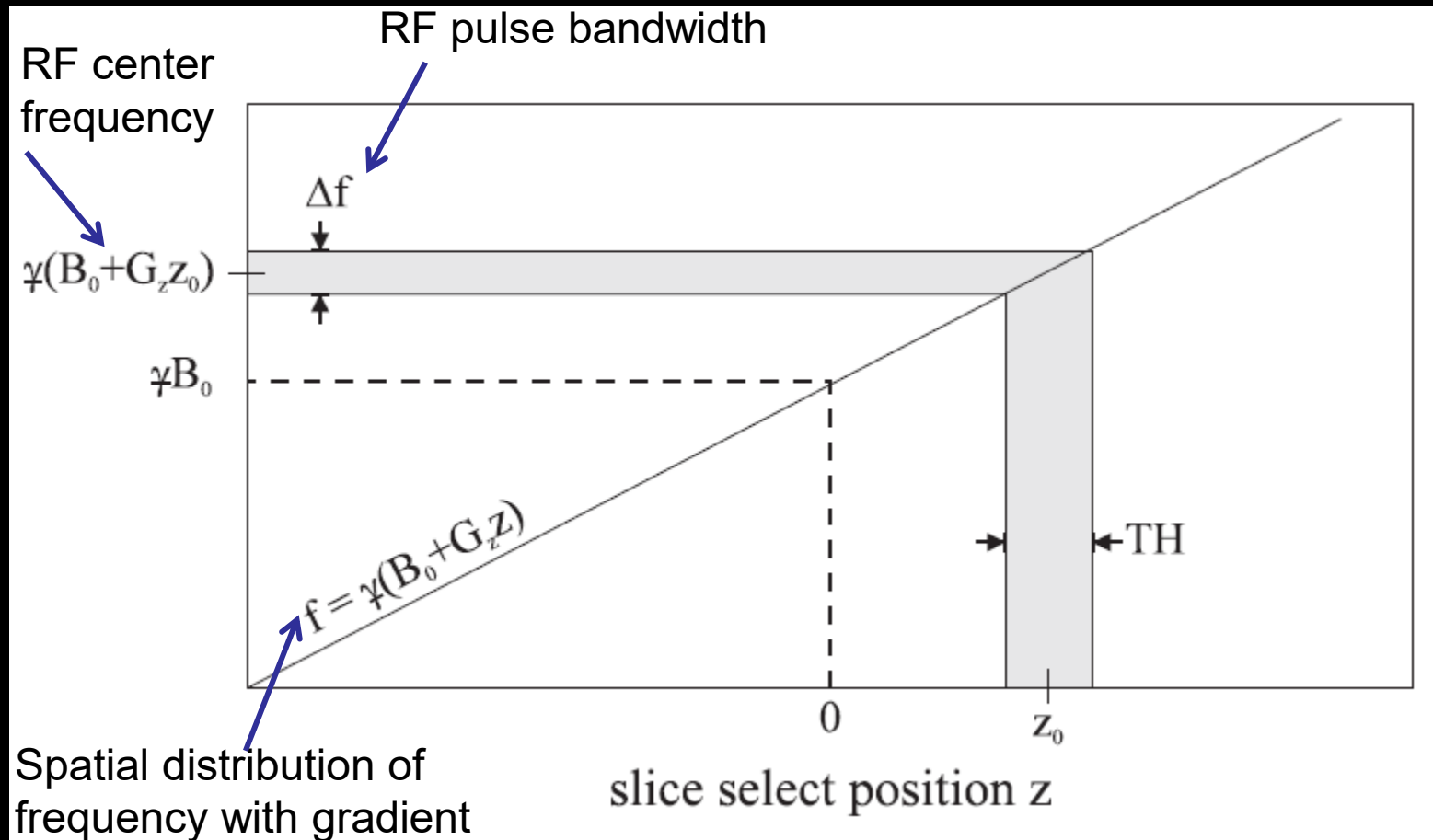
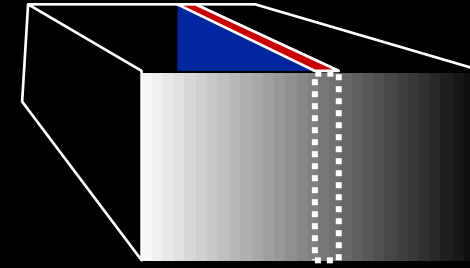


$$k_y = \gamma \cdot G_y \cdot t$$

$$= \gamma \cdot \Delta G_y \cdot \tau$$

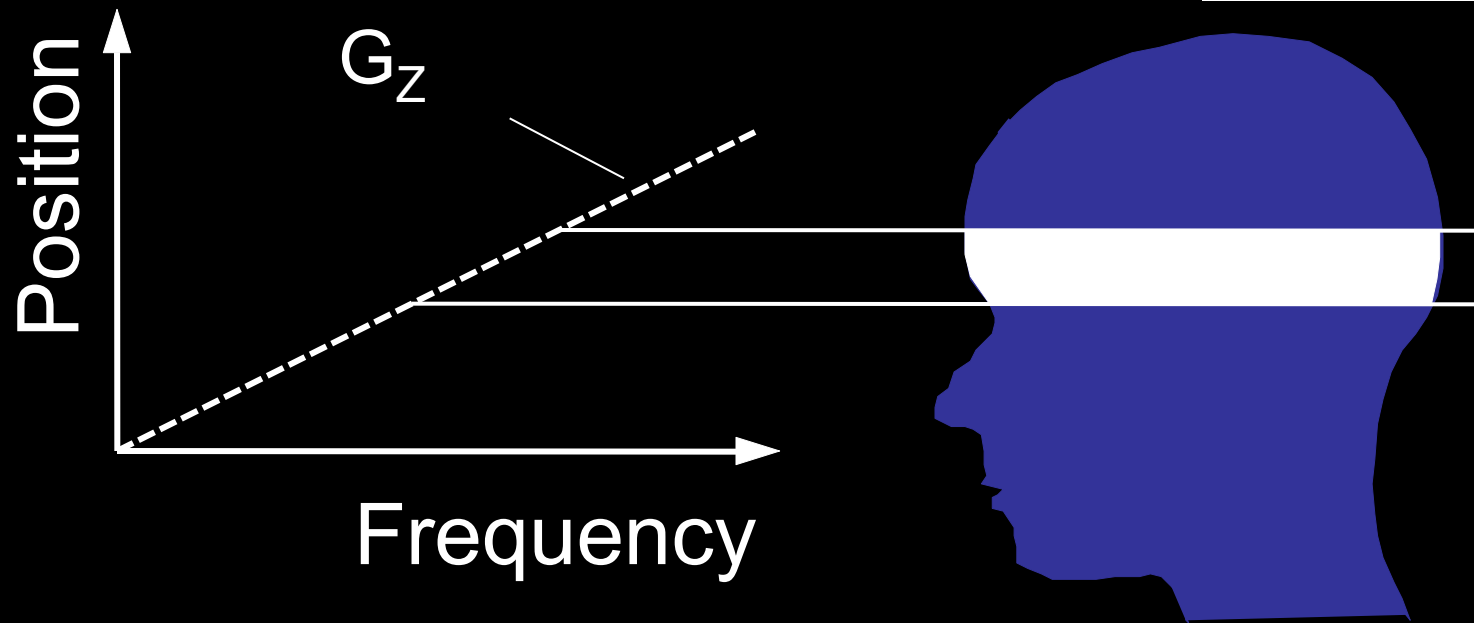
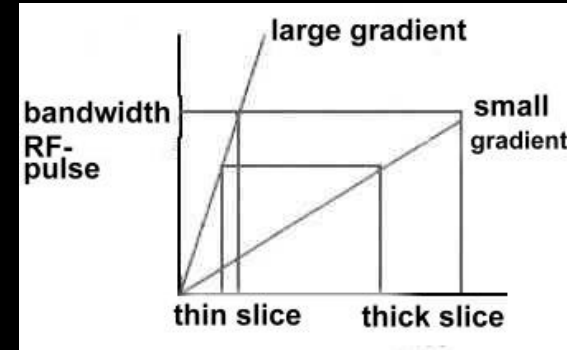


Slice selection



Note: G_{ss} can be along arbitrary direction

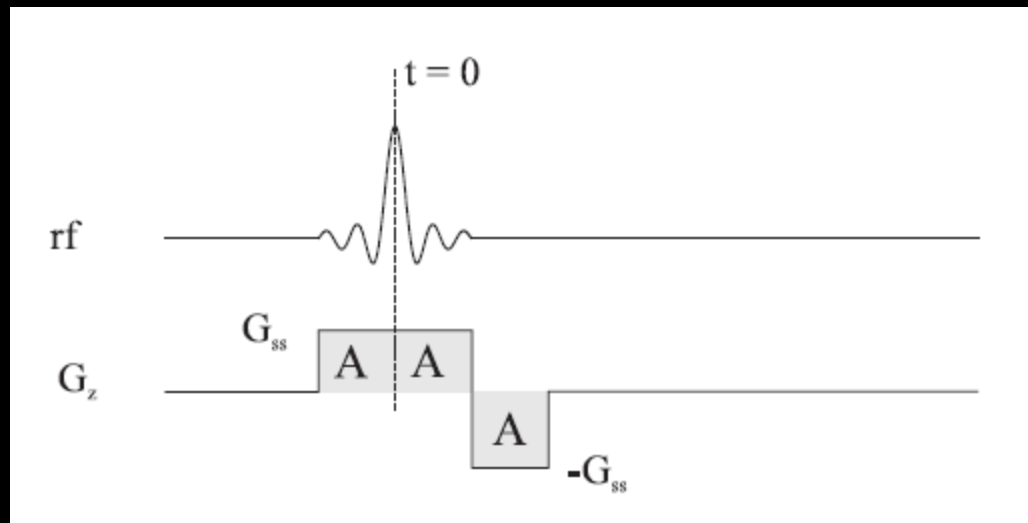
Slice Selection

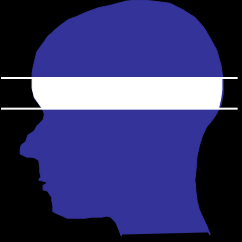


Slice selection gradient refocusing

Spin dephasing during excitation

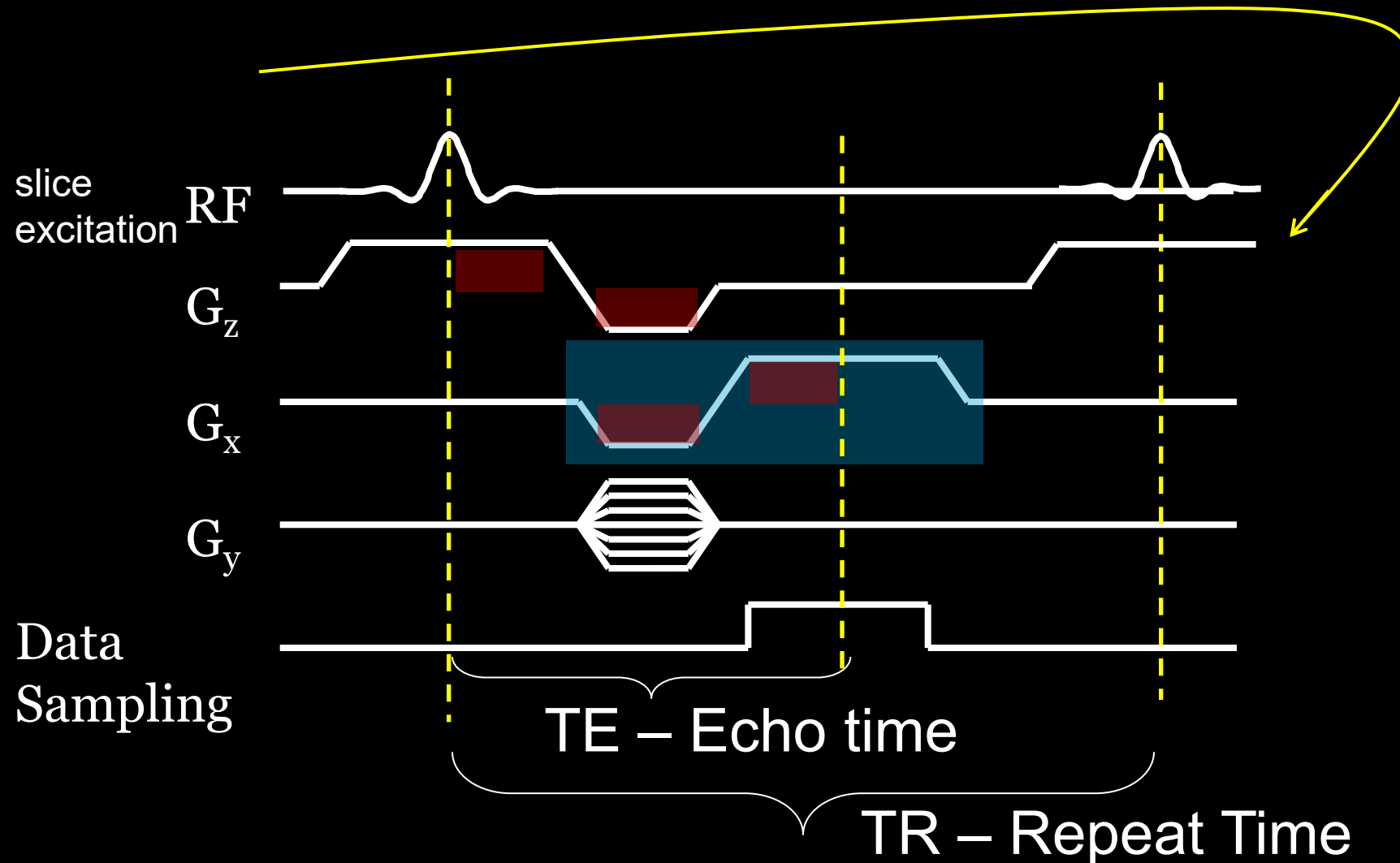
Spins are considered instantaneously tipped at the center of RF pulse.
So, they start getting dephased under the influence of G_{ss} (1/2)





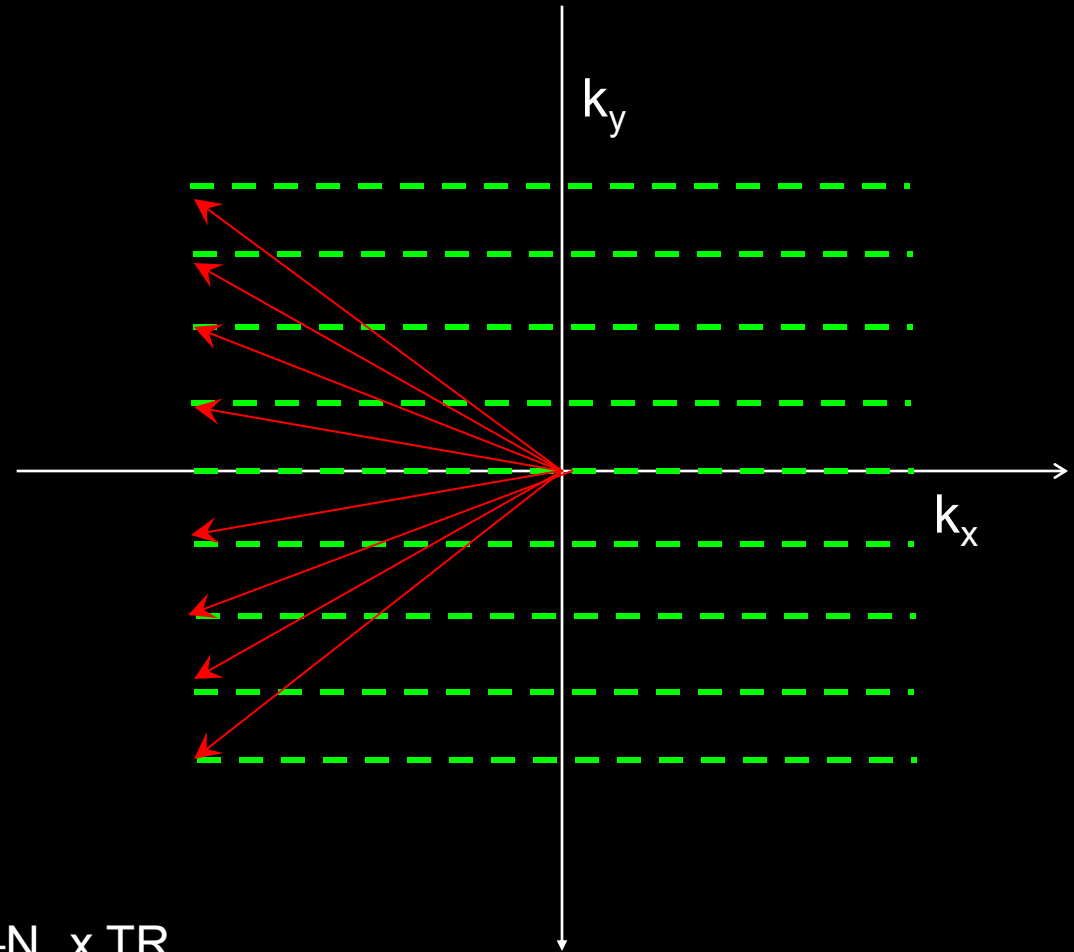
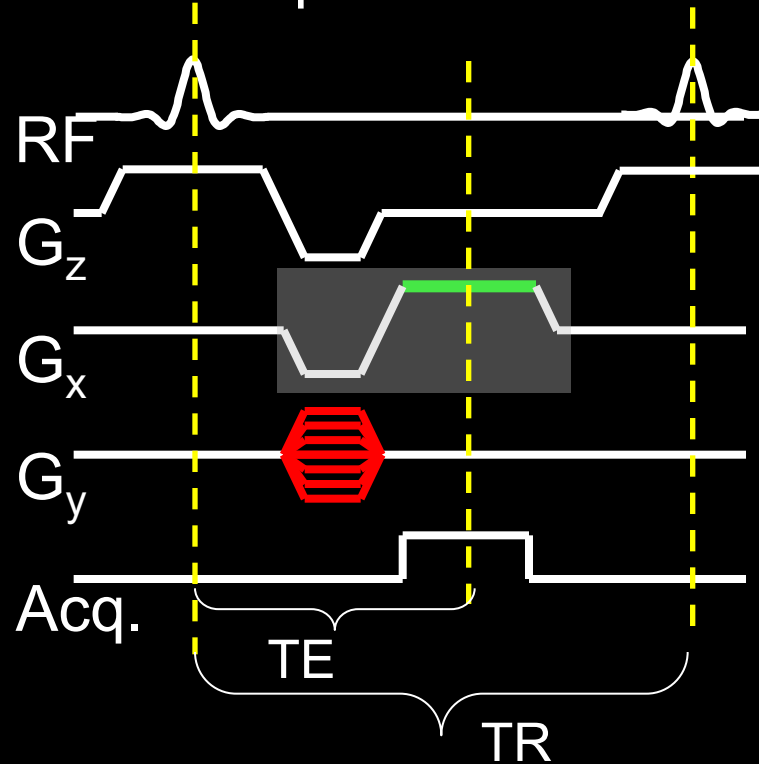
Acquiring the data... line by line

Typical **Gradient Echo** sequence



2D imaging: K-space trajectory

Typical Gradient
Echo sequence

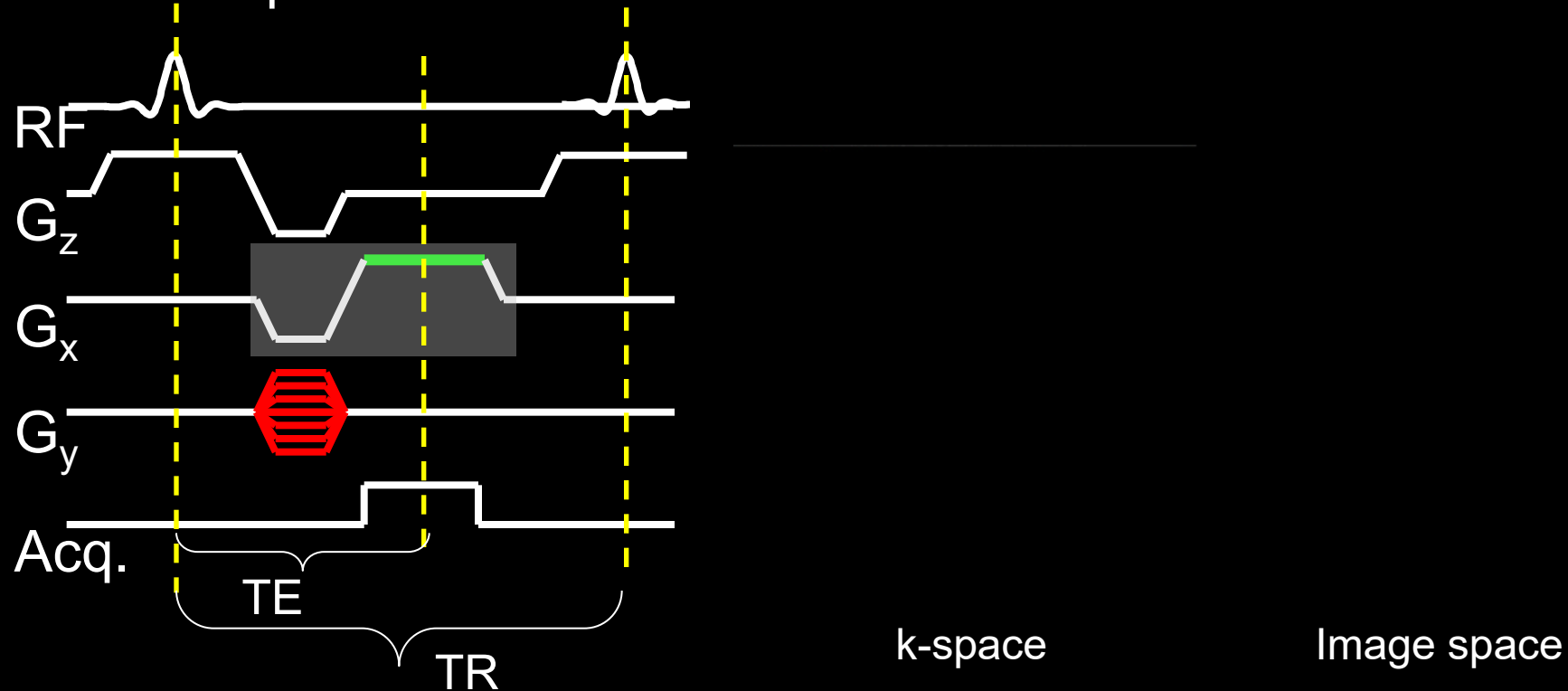


Total imaging time for a single slice $-N_y \times TR$

Sampling of the signal line by line forming the image in
Spatial frequency domain

2D imaging: K-space trajectory

Typical Gradient
Echo sequence

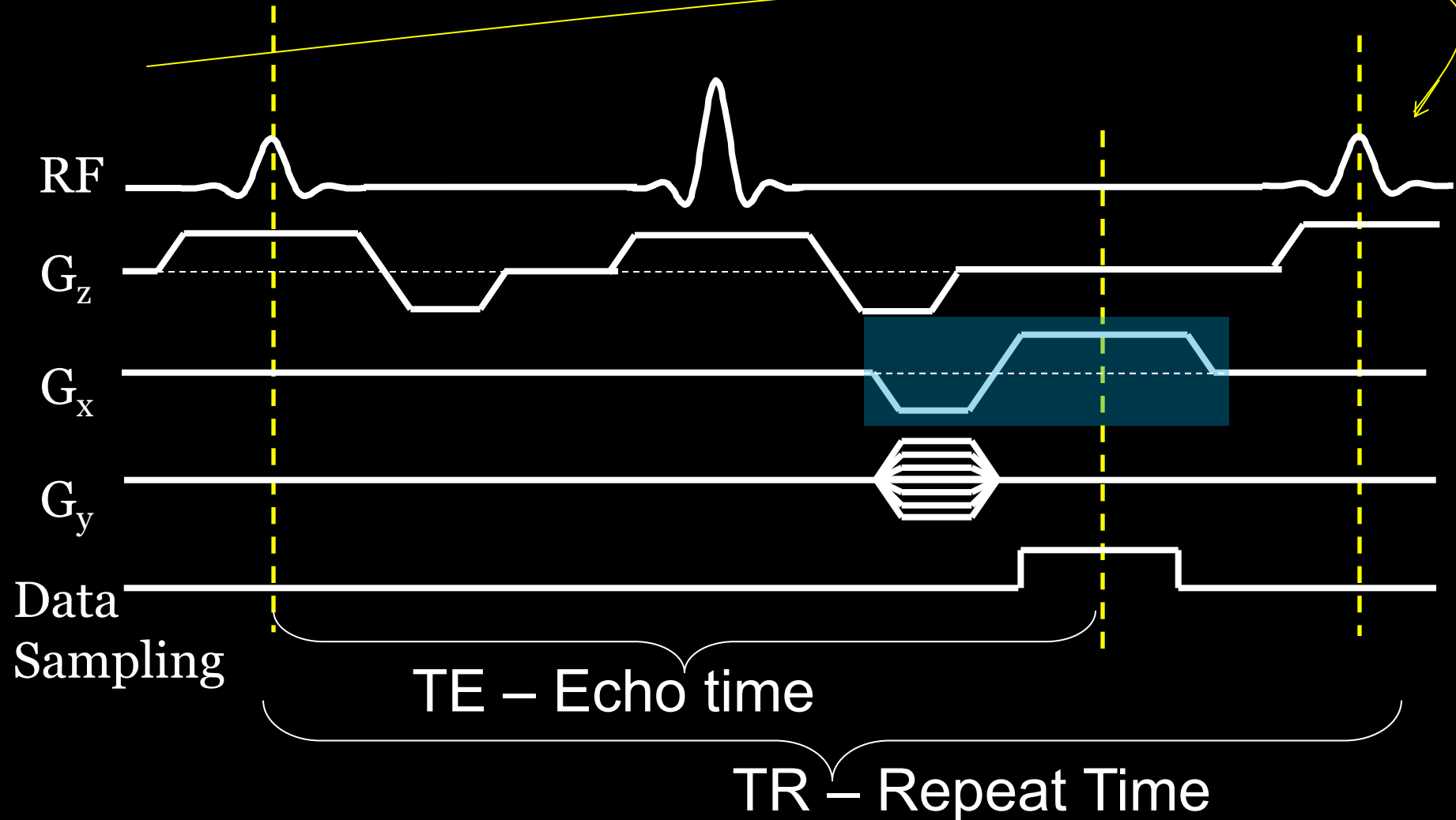


Total imaging time for a single slice $-N_y \times TR$

Sampling of the signal line by line forming the image in
Spatial frequency domain

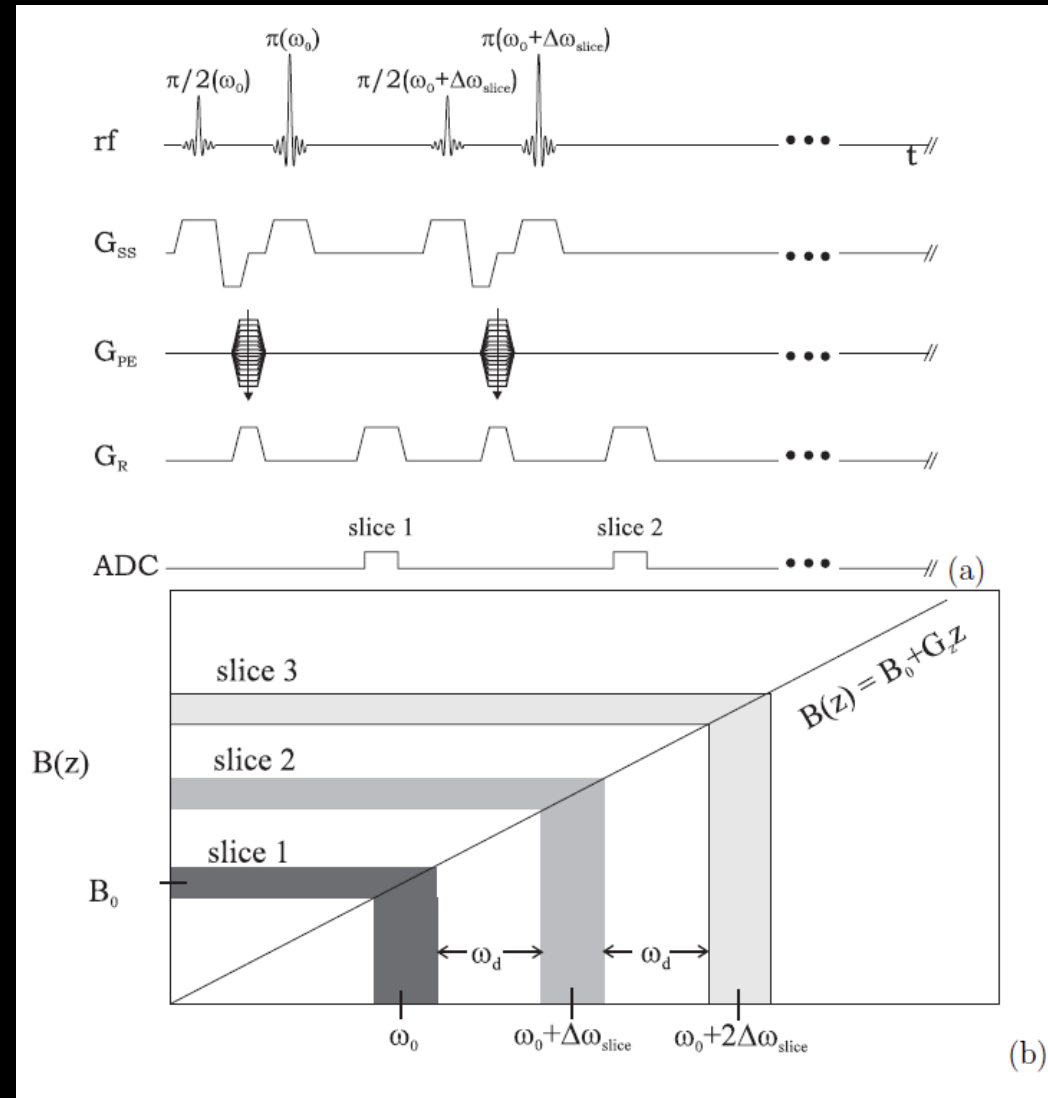
Acquiring the data... line by line

Typical **Spin Echo** sequence

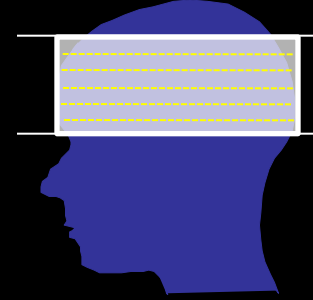


Multi-slice 2D imaging

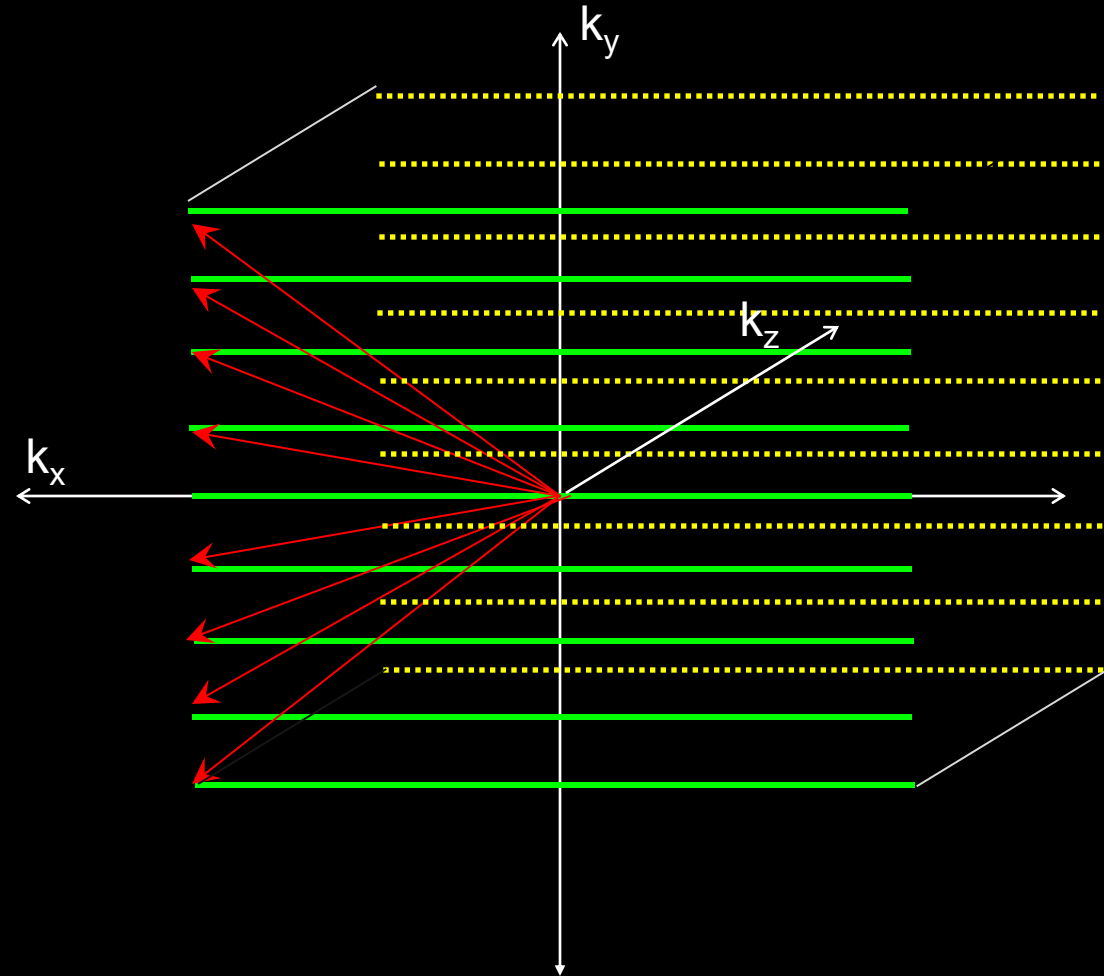
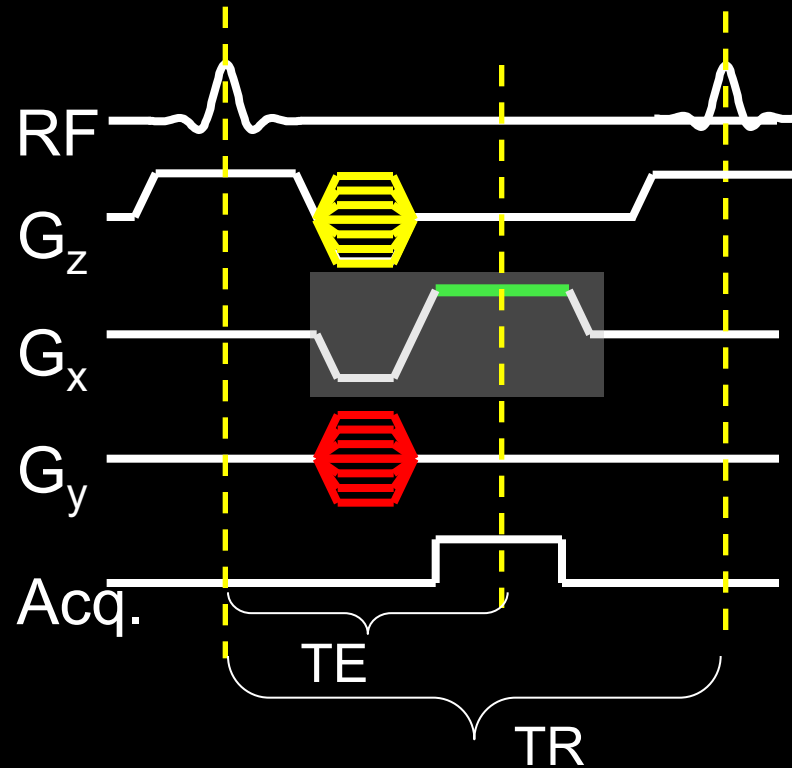
- Same G_{ss} but varying RF center frequency
- Slice gap is usually needed due to imperfect RF frequency profiles
- What is the TR here? (using the dead time)
(Hint: TR can be defined as the interval between Readouts for the same slice)



3D imaging



3D GRE - with slab excitation and partition encoding



Total Imaging time – $N_z \times N_y \times TR$

Slab Excitation and partition encoding in the z direction.

Questions